

Introduction to Global Navigation Satellite System (GNSS) Data



Themes of Day 1 of TSC

- How are the positions of a GNSS station generated? What makes up the GNSS signal?
- How do we analyze a time series of GNSS positions? How do we decompose the time series into its constituent parts?
- How do we work with a network of GNSS stations?

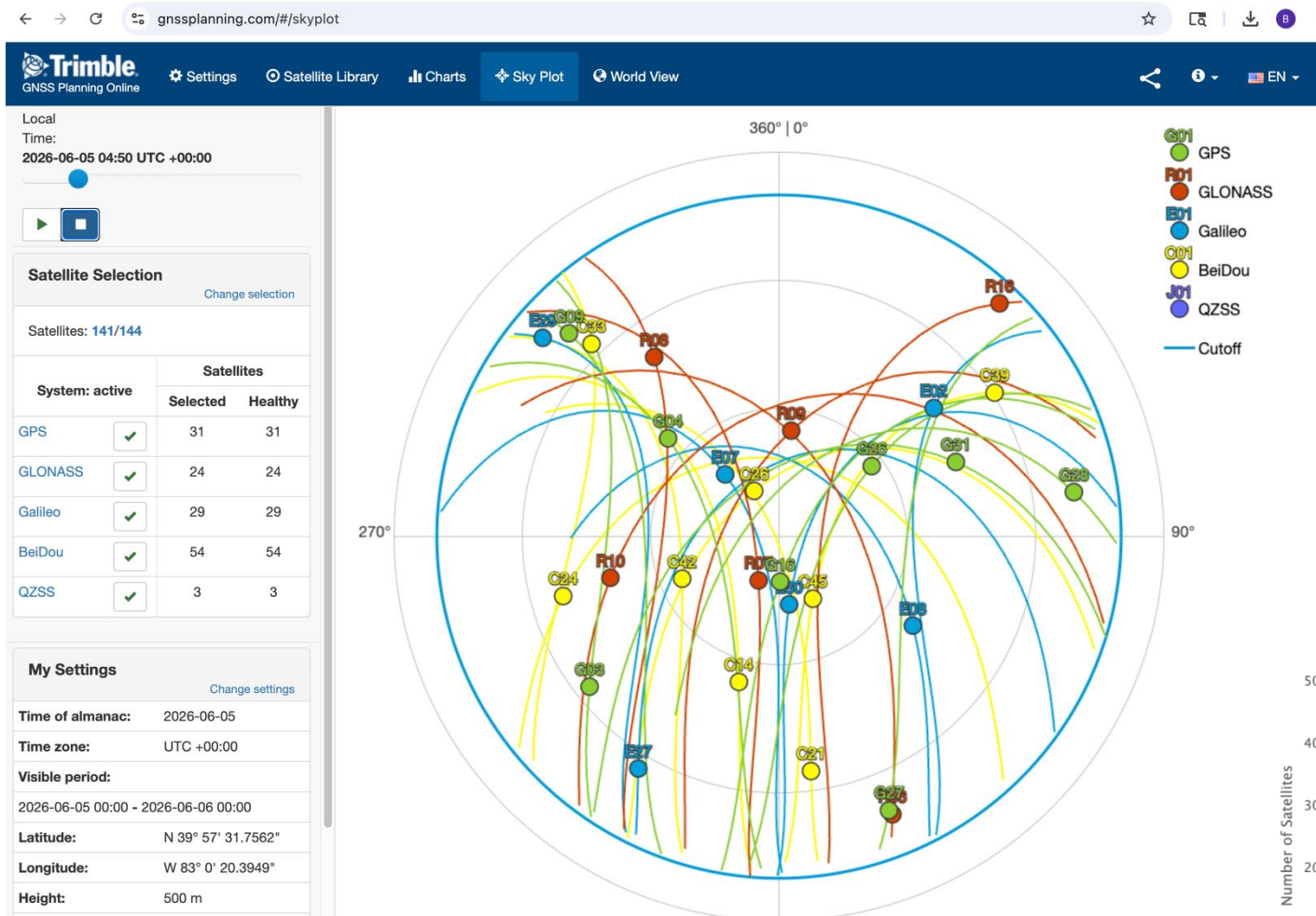
History of Satellite Navigation

- **Origins:** Cold War era navigation requirements for military aviation and missile guidance
- Both the US and USSR began development of satellite navigation systems in the early 1970s
- US program: initially called DNSS (Defense Navigation Satellite System), later renamed Navstar, then GPS
- 1983: Korean Air Flight 007 shot down over Soviet airspace. GPS subsequently opened to civilian use.

Global Navigation Satellite Systems (GNSS)

- **GPS (USA):** 24+ satellites, 6 orbital planes, 55° inclination, ~20,200 km altitude
- **GLONASS (Russia):** First launched in 1982; full global coverage restored in 2011. Satellites have slightly different frequencies.
- **BeiDou (China):** BeiDou-3 provides global coverage since 2020, earlier versions have preferential coverage over China.
- **Galileo (EU):** Civilian-focused, full constellation in 2023
- **Regional Systems:** QZSS (Japan), NavIC (India)
- **SBAS:** Satellite-Based Augmentation Systems (e.g., WAAS in USA). To improve single-frequency accuracy.

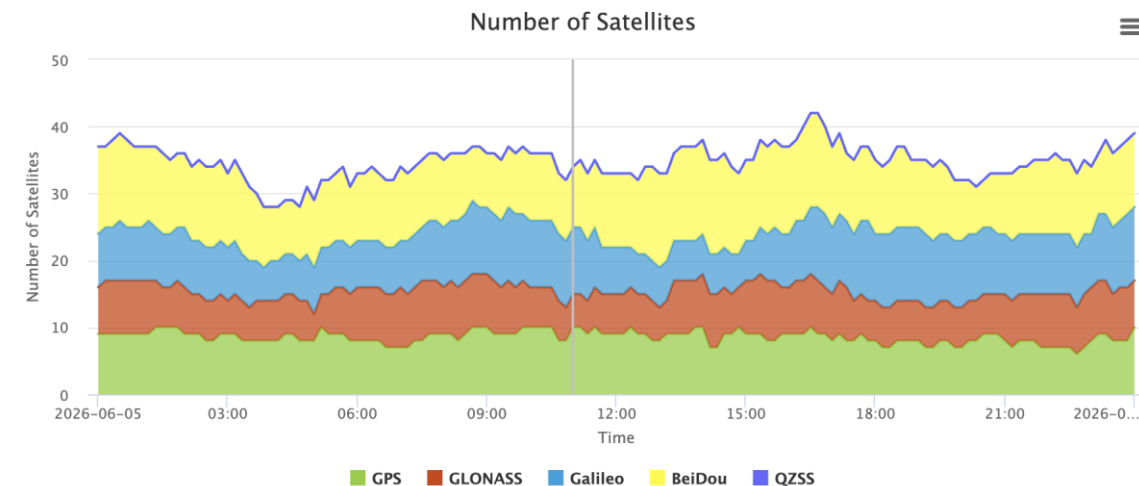
GNSS Satellites Visible



Visible satellites in Columbus last week.

Coverage hole is visible to the north due to satellite inclination.

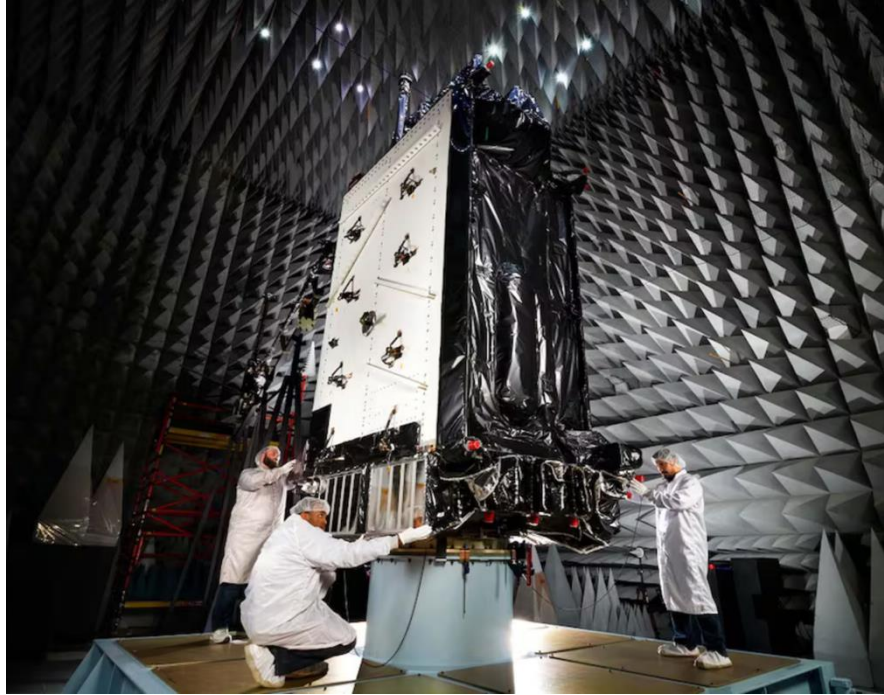
Between 30-40 satellites visible at any time.



Highcharts.com

One of my favorite websites: <https://www.gnssplanning.com/#/settings>

GPS Satellite Overview and Orbits



- GPS has 6 orbital planes at 55° inclination
- Each orbital plane has 4-5 satellites. Every location on Earth can see a minimum of 4 satellites.
- Orbital period is 1/2 sidereal day.
- There is a coverage hole at the poles where satellites are never directly overhead
- Basic information on individual satellite orbits and Pseudorandom noise (PRN) signatures is found in 'The Almanac'. This information is stored on receivers and broadcast from the satellites.

GPS is Split into Three Segments

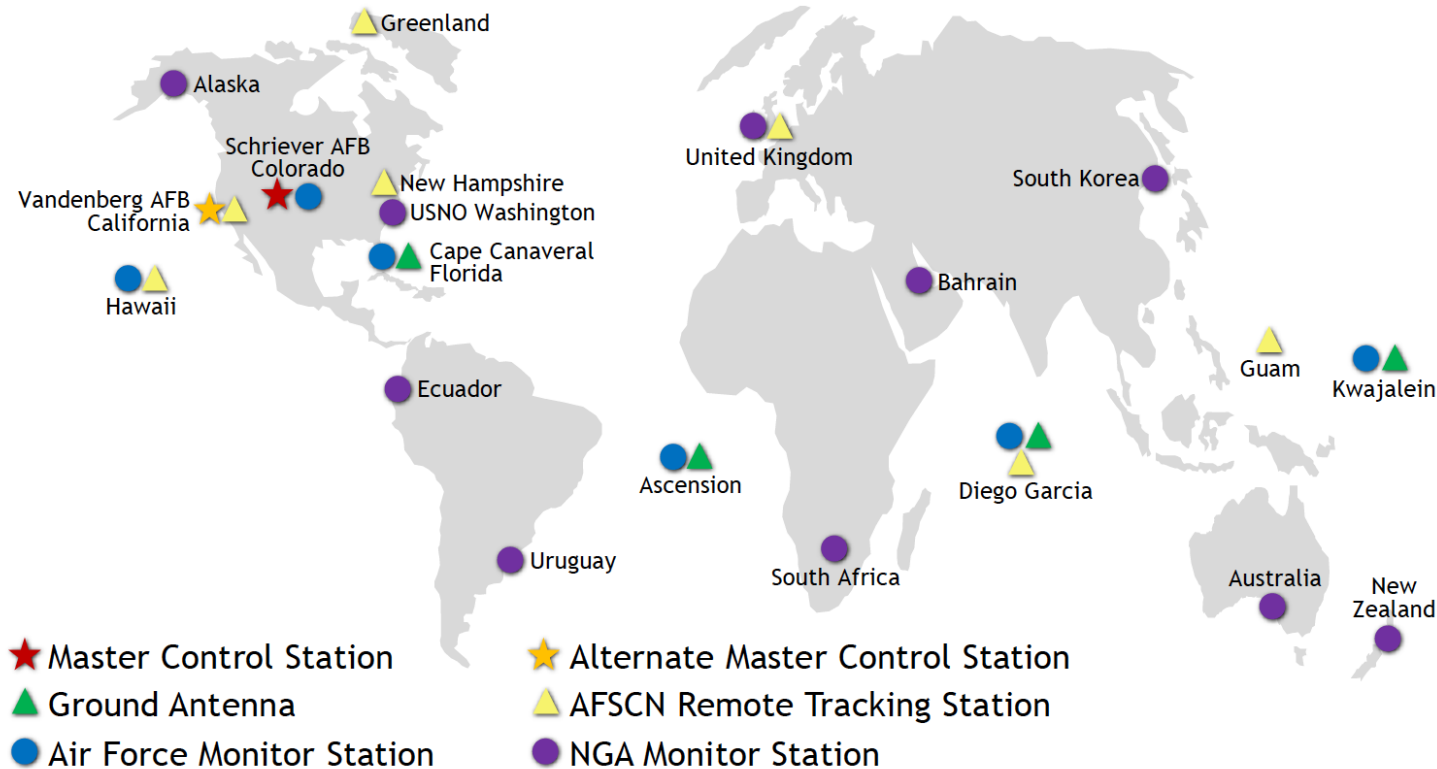
The Space Segment: This is the segment with the different GPS satellites or space vehicles (SVs). This is the part broadcasting information that you can use.

The Control Segment (or Ground): This segment is responsible for tracking the satellites (orbits and health) and providing corrections back to the satellites

The User Segment: This is the entire ecosystem of processing tools and applications that form the basis for most what scientists do with GPS.

GPS Control Segment

GPS Control Segment



Consists of Master Control Station (MCS), monitoring stations, and ground antennas.

The MCS computes all the orbital corrections needed.

The monitoring stations track each satellite and have atomic clocks for precise timing.

The ground antennas upload navigation messages and corrections to the satellites.

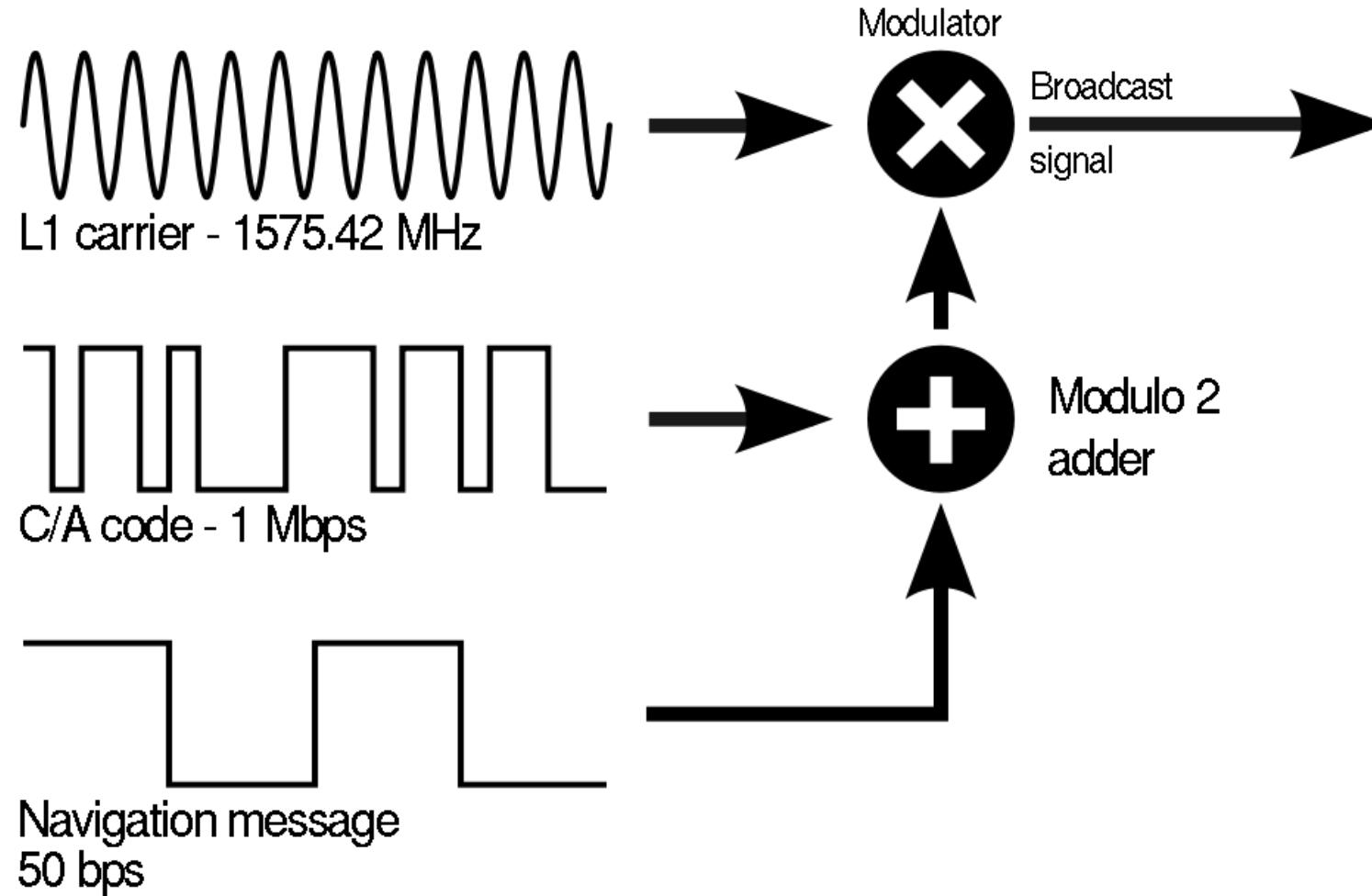
The GPS Signal

- Atomic clocks onboard the GPS satellites generate signals at a fundamental radio frequency of 10.23 MHz
- Current GPS satellites broadcast information at three **carrier** frequencies: L1, L2 and L5, which are integer multiples of the fundamental frequency (154, 120, and 115 for 1575.42 MHz, 1226.60 MHz and 1176.45 MHz).
- Each satellite also modulates a series of **pseudorandom codes** onto the carrier signal. The two that are relevant to us are the Coarse Acquisition (C/A) and precision (P) codes.

The C/A Code

- Each satellite has its own unique C/A code, the details of which are defined in the Almanac.
- The C/A code repeats itself every millisecond and is broadcast at $1/10^{\text{th}}$ of the fundamental frequency (1.023 Mbit/s). The effective wavelength of this signal is roughly 300 m.
- Embedded in the C/A code is the satellite navigation message, otherwise known as the broadcast ephemeris.
- It is a legacy code that all GPS satellites have and will transmit in the future.

The C/A Code



The P-Code

- Originally only available to military applications, but it was opened up to other users
- Is broadcast at the fundamental frequency of 10.23 MHz and repeats roughly once per week. Its effective wavelength is ten times less than the C/A code (30 m)
- The P-code is not unique to each satellite like the C/A code; an individual satellite just uses part of the master sequence.
- The P-code is broadcast on all 3 carrier frequencies.

Anti-spoofing

- The USAF modulates an additional code on top of the P-code to ensure no ground transmitter can ‘spoof’ the GPS satellites and degrade positioning service
- This is known as the W-code and is top-secret. This does add some noise to our observations, although differential GPS basically minimizes this impact
- Only DoD-approved receivers have access to the W-code
- The resulting P+W code is known as P(Y).

Selective Availability (SA)

- Prior to 2000, the USAF modulated a time varying pseudorandom code on top of the P-code, actively degrading civilian positioning
- This was officially turned off because of the use of civilian GPS instruments by the military. Once again, differential GPS was able to overcome the impacts of SA.
- While this is not used anymore, it is important to understand that it exists in older observations.

GPS Modernization

- Several modernized signals have been added to GPS satellites
- L5: Available on all GPS Block IIF satellites onward
- L2C: Adds a similar C/A code onto L2, although two different PRN values per satellite, civil moderate (CM) and civil long (CL). Block II—M onward
- L1C: A new civilian L1 code with the existing C/A code and multiplexed with additional PRN values. GPS will continue broadcasting L1 C/A in addition to L1C. Block III satellites onward
- M-code: New code separate from P(Y)

Current GPS Observation Codes

Table 10 : RINEX Version 4.00 GPS Observation Codes

GNSS System	Freq. Band / Frequency	Channel or Code	Observation Codes			
			Pseudo Range	Carrier Phase	Doppler	Signal Strength
GPS	L1/1575.42	C/A	C1C	L1C	D1C	S1C
		L1C (D)	C1S	L1S	D1S	S1S
		L1C (P)	C1L	L1L	D1L	S1L
		L1C (D+P)	C1X	L1X	D1X	S1X
		P (AS off)	C1P	L1P	D1P	S1P
		Z-tracking and similar (AS on)	C1W	L1W	D1W	S1W
		Y	C1Y	L1Y	D1Y	S1Y
		M	C1M	L1M	D1M	S1M
		codeless		L1N	D1N	S1N
	L2/1227.60	C/A	C2C	L2C	D2C	S2C
		L1(C/A) + (P2-P1) (semi-codeless)	C2D	L2D	D2D	S2D
		L2C (M)	C2S	L2S	D2S	S2S
		L2C (L)	C2L	L2L	D2L	S2L
		L2C (M+L)	C2X	L2X	D2X	S2X
		P (AS off)	C2P	L2P	D2P	S2P
		Z-tracking and similar (AS on)	C2W	L2W	D2W	S2W
		Y	C2Y	L2Y	D2Y	S2Y
		M	C2M	L2M	D2M	S2M
		codeless		L2N	D2N	S2N
	L5/1176.45	I	C5I	L5I	D5I	S5I
		Q	C5Q	L5Q	D5Q	S5Q
		I+Q	C5X	L5X	D5X	S5X

The GPS system has 3 primary frequencies, but your receiver is recording specific multiplexed versions of code observations.

The other GNSS systems have a similar makeup but with different frequencies and code definitions.

See the RINEX4 Manual for more details
https://files.igs.org/pub/data/format/rinex_4.00.pdf

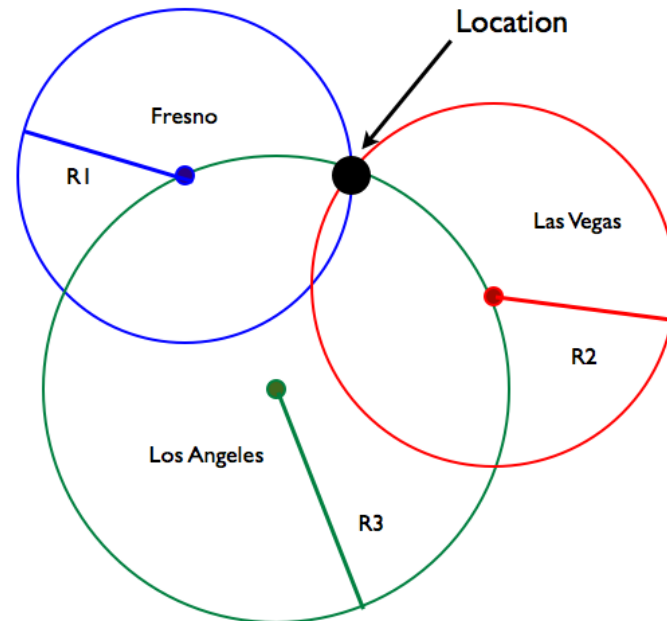
Basic Positioning

Ranging and Trilateration

The range between two objects is defined as the Euclidean distance between those objects:

$$\text{Range} = \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}$$

Given three range observations, the location of a point can be found through trilateration



This is identical to the problem of earthquake location with P wave arrival times. Since there are 4 unknowns (hypocenter plus origin time), we would need 4 stations (or satellites for positioning).

Pseudorange

In GPS, we calculate the same range parameter except embedded in the range are other known/unknown path delays, errors, and biases, which is the **pseudorange**:

$$Pseudorange = Range + errors$$

What we observe is the timing at the satellite and the receiver. The receiver matches the PRN code sequence with the known PRN and the associated time stamp. What we get looks like:

$$Pseudorange = c(T_r - T_s)$$

Where c is the speed of light

Pseudorange: Clock Biases

Each clock has a bias associated with it. The receiver clock bias is usually much larger since they use quartz oscillators (up to 1 s errors) rather than Cesium-Rubidium clocks (1 ns errors) on the satellites. Note, the clocks on GPS satellites run at a slightly slower rate to counteract relativity.

Each recorded time will be the sum of the true time and the clock bias such that:

$$T_r = t_r + \tau_r$$

$$T_s = t_s + \tau_s$$

So, the pseudorange will be:

$$\textit{Pseudorange} = c(t_r + \tau_r - t_s - \tau_s)$$

Pseudorange: Clock Biases

The **true range** is defined by the '**true times**' at the receiver and satellite such that:

$$\text{range}(t_r, t_s) = \rho(t_r, t_s) = c(t_r - t_s)$$

We can rewrite the pseudorange equation as:

$$P(t_r, t_s) = \rho(t_r, t_s) + c(\tau_r - \tau_s)$$

$$\rho(t_r, t_s) = \sqrt{(x_r(t_r) - x_s(t_s))^2 + (y_r(t_r) - y_s(t_s))^2 + (z_r(t_r) - z_s(t_s))^2}$$

In this equation, the range will give us the Cartesian coordinates of the station we want to solve for. We also need to be careful when computing satellite location that we use the **satellite transmit time** and not the **receiver time**.

Pseudorange Observables

Each GPS receiver will try to sync up the code observations (C/A and P-code) with the PRN sequence associated with the satellite

What the receiver measures is the time difference, $(T_r - T_s)$

In our current incarnation of GPS, this will result in three pseudorange observations:

C/A on L1 = known as C1

P on L1 = known as P1

P on L2 – known as P2

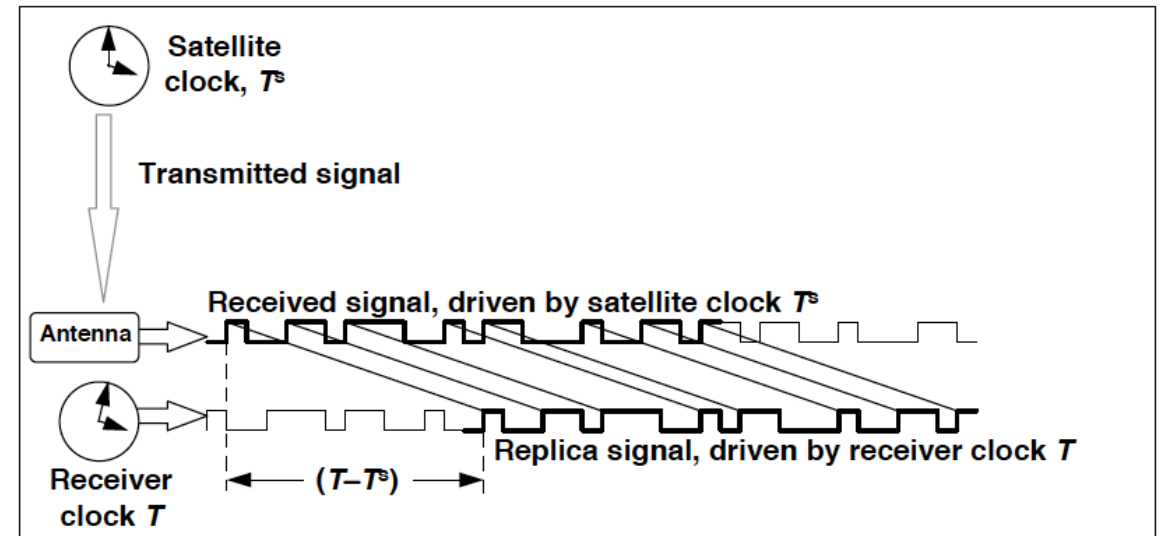


Figure 1: A schematic diagram showing how the GPS pseudorange observation is related to the satellite and receiver clocks.

Point Positioning with Pseudorange:

Let's assume we know the satellite orbits and clock biases well.

The bare minimum number of satellites to compute your receiver position is 4.
Here, subscripts indicate satellite number and xyz are the receiver coordinates:

$$P_1 = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2} + c\tau_r - c\tau_1$$

$$P_2 = \sqrt{(x - x_2)^2 + (y - y_2)^2 + (z - z_2)^2} + c\tau_r - c\tau_2$$

$$P_3 = \sqrt{(x - x_3)^2 + (y - y_3)^2 + (z - z_3)^2} + c\tau_r - c\tau_3$$

$$P_4 = \sqrt{(x - x_4)^2 + (y - y_4)^2 + (z - z_4)^2} + c\tau_r - c\tau_4$$

To solve this, we need to perform a Taylor's expansion. First, let us define the following:

$$P_{obs} = P_{model} + noise$$

$$P_{obs} = P(x, y, z, \tau_r) + v$$

Point Positioning with Pseudorange:

First, Taylor's expansion:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots$$

If we ignore second order and beyond terms, we get:

$$P(x, y, z, \tau) = P(x_0, y_0, z_0, \tau_0) + (x - x_0) \frac{\partial P}{\partial x} + (y - y_0) \frac{\partial P}{\partial y} + (z - z_0) \frac{\partial P}{\partial z} + (\tau - \tau_0) \frac{\partial P}{\partial \tau}$$

The subscript **0** terms indicates an '**a priori**' receiver coordinate and clock bias.

This is then rewritten as:

$$P_{obs} = P_{computed} + \Delta x \frac{\partial P}{\partial x} + \Delta y \frac{\partial P}{\partial y} + \Delta z \frac{\partial P}{\partial z} + \Delta \tau \frac{\partial P}{\partial \tau} + v \quad \Delta P = \left(\frac{\partial P}{\partial x}, \frac{\partial P}{\partial y}, \frac{\partial P}{\partial z}, \frac{\partial P}{\partial \tau} \right) \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \\ \Delta \tau \end{pmatrix} + v$$

Point Positioning with Pseudorange:

Starting with the pseudorange distance difference equation:

$$\Delta P = \begin{pmatrix} \frac{\partial P}{\partial x}, \frac{\partial P}{\partial y}, \frac{\partial P}{\partial z}, \frac{\partial P}{\partial \tau} \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \\ \Delta \tau \end{pmatrix} + v$$

We then expand this into matrix form for the 4 satellites:

$$\begin{pmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \end{pmatrix} = \begin{pmatrix} \frac{\partial P_1}{\partial x} & \frac{\partial P_1}{\partial y} & \frac{\partial P_1}{\partial z} & \frac{\partial P_1}{\partial \tau} \\ \frac{\partial P_2}{\partial x} & \frac{\partial P_2}{\partial y} & \frac{\partial P_2}{\partial z} & \frac{\partial P_2}{\partial \tau} \\ \frac{\partial P_3}{\partial x} & \frac{\partial P_3}{\partial y} & \frac{\partial P_3}{\partial z} & \frac{\partial P_3}{\partial \tau} \\ \frac{\partial P_4}{\partial x} & \frac{\partial P_4}{\partial y} & \frac{\partial P_4}{\partial z} & \frac{\partial P_4}{\partial \tau} \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \\ \Delta \tau \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

$$b = Ax + v$$

Point Positioning with Pseudorange:

These terms are purely a function of direction to each satellite (direction cosines)

$$\begin{pmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \end{pmatrix} = \begin{pmatrix} \frac{\partial P_1}{\partial x} & \frac{\partial P_1}{\partial y} & \frac{\partial P_1}{\partial z} & \frac{\partial P_1}{\partial \tau} \\ \frac{\partial P_2}{\partial x} & \frac{\partial P_2}{\partial y} & \frac{\partial P_2}{\partial z} & \frac{\partial P_2}{\partial \tau} \\ \frac{\partial P_3}{\partial x} & \frac{\partial P_3}{\partial y} & \frac{\partial P_3}{\partial z} & \frac{\partial P_3}{\partial \tau} \\ \frac{\partial P_4}{\partial x} & \frac{\partial P_4}{\partial y} & \frac{\partial P_4}{\partial z} & \frac{\partial P_4}{\partial \tau} \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \\ \Delta \tau \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

This is simply the speed of light, c

Taking the partial derivatives of the pseudorange equation gives us this form

$$\begin{pmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \end{pmatrix} = \begin{pmatrix} \frac{x_0 - x_{s1}}{\rho_1} & \frac{y_0 - y_{s1}}{\rho_1} & \frac{z_0 - z_{s1}}{\rho_1} & c \\ \frac{x_0 - x_{s2}}{\rho_2} & \frac{y_0 - y_{s2}}{\rho_2} & \frac{z_0 - z_{s2}}{\rho_2} & c \\ \frac{x_0 - x_{s3}}{\rho_3} & \frac{y_0 - y_{s3}}{\rho_3} & \frac{z_0 - z_{s3}}{\rho_3} & c \\ \frac{x_0 - x_{s4}}{\rho_4} & \frac{y_0 - y_{s4}}{\rho_4} & \frac{z_0 - z_{s4}}{\rho_4} & c \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \\ \Delta \tau \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

Note how I use subscript s_1, s_2, s_3, s_4 to differentiate from the *a priori* receiver location x_0

Point Positioning with Pseudorange

Everything in the design matrix is known, so this problem can be solved with least squares. The estimated terms are the x, y, z displacements with respect to the a priori coordinates and the receiver clock bias.

$$\begin{pmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \end{pmatrix} = \begin{pmatrix} \frac{x_0 - x_{s1}}{\rho_1} & \frac{y_0 - y_{s1}}{\rho_1} & \frac{z_0 - z_{s1}}{\rho_1} & c \\ \frac{x_0 - x_{s2}}{\rho_2} & \frac{y_0 - y_{s2}}{\rho_2} & \frac{z_0 - z_{s2}}{\rho_2} & c \\ \frac{x_0 - x_{s3}}{\rho_3} & \frac{y_0 - y_{s3}}{\rho_3} & \frac{z_0 - z_{s3}}{\rho_3} & c \\ \frac{x_0 - x_{s4}}{\rho_4} & \frac{y_0 - y_{s4}}{\rho_4} & \frac{z_0 - z_{s4}}{\rho_4} & c \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \\ \Delta \tau \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

Pseudorange Dilution of Precision

- The dilution of precision (DOP) is a way to assess how good your constellation geometry is.
- This can be computed well in advance since it is only dependent on the design matrix, A, from the previous slide.
- The covariance is related to the observation error and the cofactor matrix (A) by the following:

$$C = \sigma^2 (A^T A)^{-1} = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{xz} & \sigma_{x\tau} \\ \sigma_{yx} & \sigma_y^2 & \sigma_{yz} & \sigma_{y\tau} \\ \sigma_{zx} & \sigma_{zy} & \sigma_z^2 & \sigma_{z\tau} \\ \sigma_{\tau x} & \sigma_{\tau y} & \sigma_{\tau z} & \sigma_\tau^2 \end{pmatrix}$$

Pseudorange Dilution of Precision

$$VDOP = \sigma_u$$

$$HDOP = \sqrt{\sigma_n^2 + \sigma_e^2}$$

$$PDOP = \sqrt{\sigma_n^2 + \sigma_e^2 + \sigma_u^2}$$

$$TDOP = \sigma_\tau$$

$$GDOP = \sqrt{\sigma_n^2 + \sigma_e^2 + \sigma_u^2 + \sigma_\tau^2}$$

There are 5 dilution of precision (DOP) parameters (after rotating into local coordinates):

Good
DOP



Bad
DOP



DOP Value	Rating ^[5]	Description
< 1	Ideal	Highest possible confidence level to be used for applications demanding the highest possible precision at all times.
1–2	Excellent	At this confidence level, positional measurements are considered accurate enough to meet all but the most sensitive applications.
2–5	Good	Represents a level that marks the minimum appropriate for making accurate decisions. Positional measurements could be used to make reliable in-route navigation suggestions to the user.
5–10	Moderate	Positional measurements could be used for calculations, but the fix quality could still be improved. A more open view of the sky is recommended.
10–20	Fair	Represents a low confidence level. Positional measurements should be discarded or used only to indicate a very rough estimate of the current location.
> 20	Poor	At this level, measurements should be discarded.

Phase Observables

For geodetic quality positioning, we take advantage of the phase of the GPS carrier signals. The accuracy is reduced to under 2 cm for horizontal components (as opposed to 1-10 meters for code-based positioning).

Phase can be linked to time and frequency by the following:

$$T(t) = \frac{\phi(t) - \phi_0}{f_0}$$

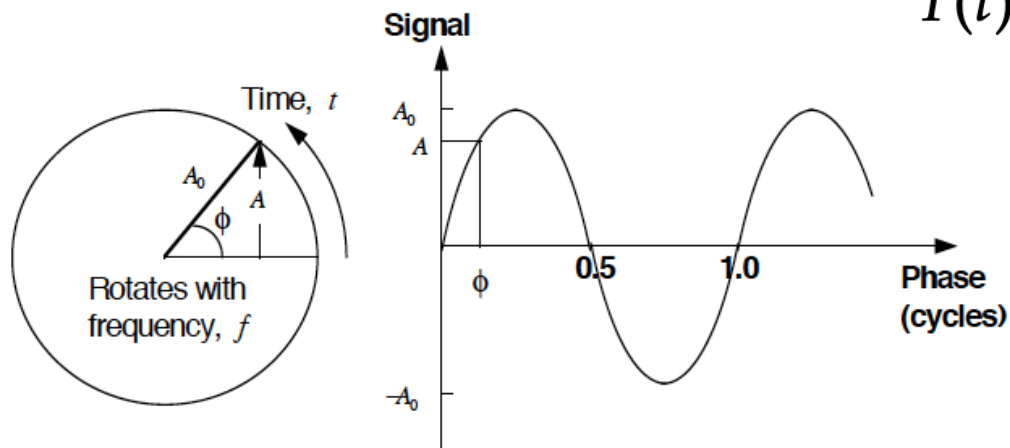


Figure 2: The meaning of phase.

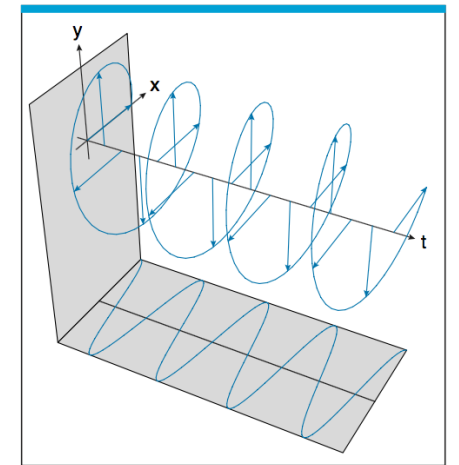


Figure 1. At a fixed point in space, the electric field vector of a right-hand circularly polarized wave rotates clockwise as seen from the wave's source.

Phase Observables: Beat Phase

The recorded signal is the carrier phase modulated by the PRN code, which makes it difficult to record the phase.

The secret is to multiply the GPS signal ($G(t)$) with a replica carrier signal ($R(t)$):

$$\begin{aligned} R(t) \otimes G(t) &= R_0 \sin(2\pi\phi_R(t)) \times G_0 \sin(2\pi\phi_G(t)) \\ &= \frac{R_0 G_0}{2} [\cos 2\pi(\phi_R(t) - \phi_G(t)) - \cos 2\pi(\phi_R(t) + \phi_G(t))] \end{aligned}$$

There is both a high and low frequency component. High pass filtering removes the right side of the cosines. The result is the **carrier beat phase**:

$$B(t) = \frac{R_0 G_0}{2} \cos 2\pi(\phi_R(t) - \phi_G(t)) = B_0 \cos 2\pi\phi_B(t)$$

Phase Observables: Beat Phase

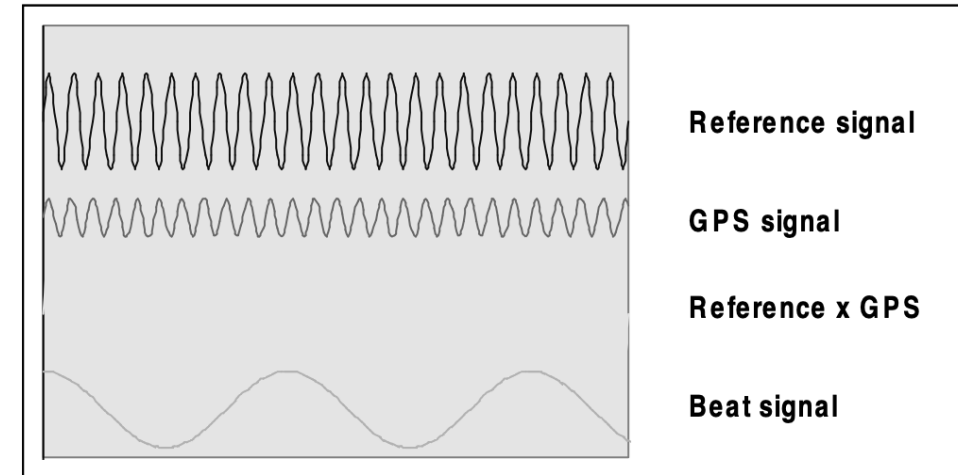
The beat phase and frequencies are just the difference between the receiver and observed signals:

$$\phi_B(t) = \phi_R(t) - \phi_G(t)$$
$$f_B = f_R - f_G$$

Now that we are measuring the phase in cycles, we have one major problem: we do not know which cycle we are on since they are constantly repeating.

If we record phase, Φ , we will have some integer number of cycles, N , such that:

$$\Phi + N = \phi_R - \phi_G$$



Phase Observables

Going from our general equation, we can rewrite things a bit:

$$\Phi + N = \phi_R - \phi_G$$

We first substitute the following for the phase terms with clock times:

$$\phi(T) = f_0 T + \phi_0$$

$$\phi_{transmit}^s(T^s) = f_0 T_{transmit}^s + \phi_0^s$$

And substituting into the top equation yields:

$$\Phi + N = f_0 T + \phi_0 - (f_0 T_{transmit}^s + \phi_0^s)$$

$$\Phi^s(T) = f_0 (T - T^s) + \phi_0 - \phi_0^s - N^s$$

Phase Observables to Range

If we multiply the phase observation by the nominal wavelength of the carrier signal, we obtain the range, L :

$$\Phi^s(T) = f_0(T - T^s) + \phi_0 - \phi_0^s - N^s$$

$$L_r^s(T_r) = \lambda_0 \Phi_r^s(T_r)$$

$$L_r^s(T_r) = \lambda_0 f_0(T_r - T^s) + \lambda_0(\phi_{0r} - \phi_0^s - N_r^s)$$

$$L_r^s(T_r) = c(T_r - T^s) + B_r^s$$

Notice this is the same as the pseudorange equation except the carrier phase bias is rolled into a single parameter, **B**. Within **B** are **integer phase ambiguities** and **fractional cycle biases** (FCBs).

GPS Orbits

- Accurate orbits are needed to compute the true range from satellite to receiver.
- Every GPS satellite broadcasts its navigation parameters on top of the C/A code (on L1) at a bitrate of 50 bits/s.
- Broadcast messages are updated every two hours and are usable up to 4 hours total (2 hours before, 2 hours after). After this, the orbital extrapolation errors become large.
- The orbital parameters are often called the ‘**ephemeris**’ or ‘**ephemerides**’ for plural. In GPS, the broadcast orbits are Keplerian.
- The broadcast ephemeris is generally accurate to 1 meter

Precise Orbits

Since the broadcast ephemeris is only accurate to ~1 meter, there is a need in the geodetic community for precise orbits

Many organizations generate precise orbits, the main ones are the IGS (International GNSS Service), CODE (Center for Orbit Determination in Europe), GFZ, ESA, Wuhan, and JPL.

The three main products released are: **ultra-rapid, rapid, and final orbits**

In addition to the orbital information, they include satellite clock corrections as well

Precise Orbits

GPS Satellite Ephemerides / Satellite & Station Clocks

Type		Accuracy	Latency	Updates	Sample Interval
Broadcast	orbits	~100 cm	real time	--	daily
	Sat. clocks	~5 ns RMS ~2.5 ns SDev			
Ultra-Rapid (predicted half)	orbits	~5 cm	real time	at 03, 09, 15, 21 UTC	15 min
	Sat. clocks	~3 ns RMS ~1.5 ns SDev			
Ultra-Rapid (observed half)	orbits	~3 cm	3 - 9 hours	at 03, 09, 15, 21 UTC	15 min
	Sat. clocks	~150 ps RMS ~50 ps SDev			
Rapid	orbits	~2.5 cm	17 - 41 hours	at 17 UTC daily	15 min
	Sat. & Stn. clocks	~75 ps RMS ~25 ps SDev			5 min
Final	orbits	~2.5 cm	12 - 18 days	every Thursday	15 min
	Sat. & Stn. clocks	~75 ps RMS ~20 ps SDev			Sat.: 30s Stn.: 5 min

Orbit Calculations

Computing orbits for positioning is beyond this course, but I have examples for broadcast and precise orbits here:

https://github.com/crowellbw/SNIVEL/blob/master/SNIVEL_orbits.py

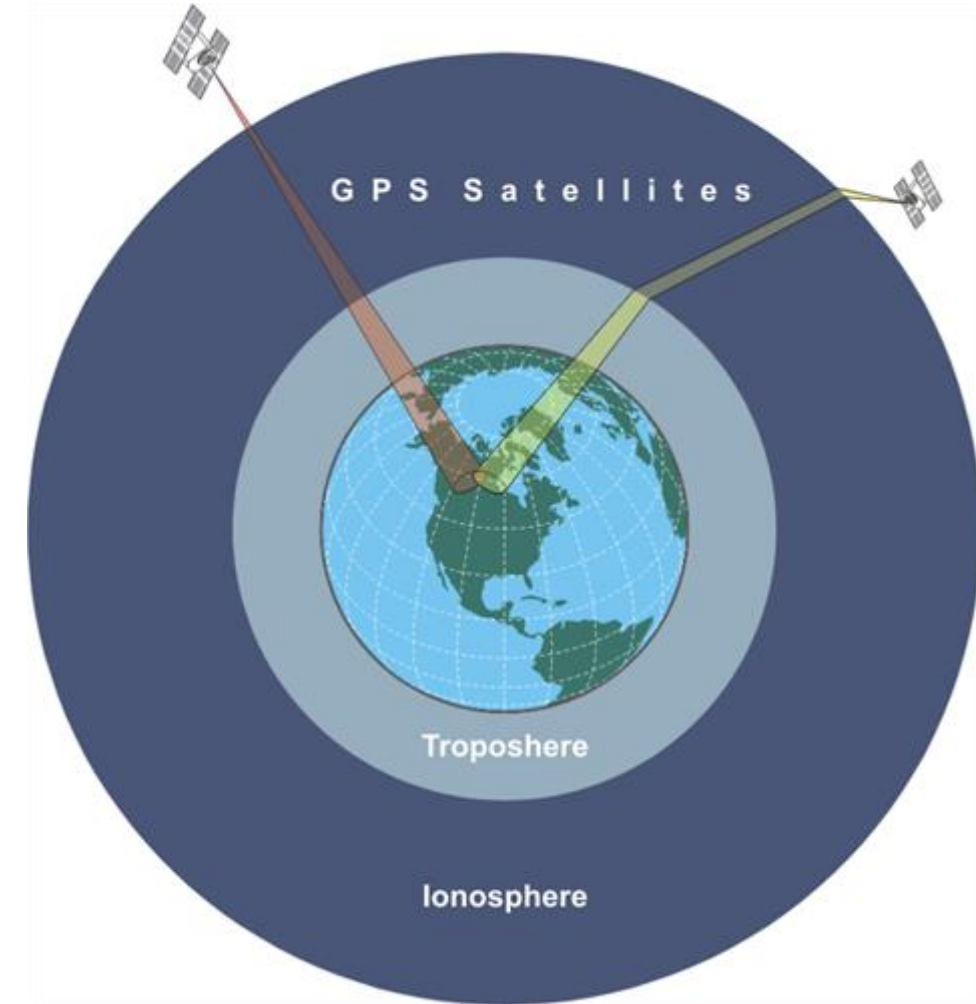
For broadcast, Keplerian parameters are provided every 2 hours. Using these values, the orbital position at the satellite broadcast is calculated (https://gssc.esa.int/navipedia/index.php?title=GPS_and_Galileo_Satellite_Coordinates_Computation)

For precise orbits, a Lagrangian interpolation is performed to get the orbital position at the broadcast time. The orbits are provided as ITRF Cartesian positions at a cadence of 5 to 15 minutes. Satellite clocks are treated similarly.

Atmospheric and Ionospheric Corrections

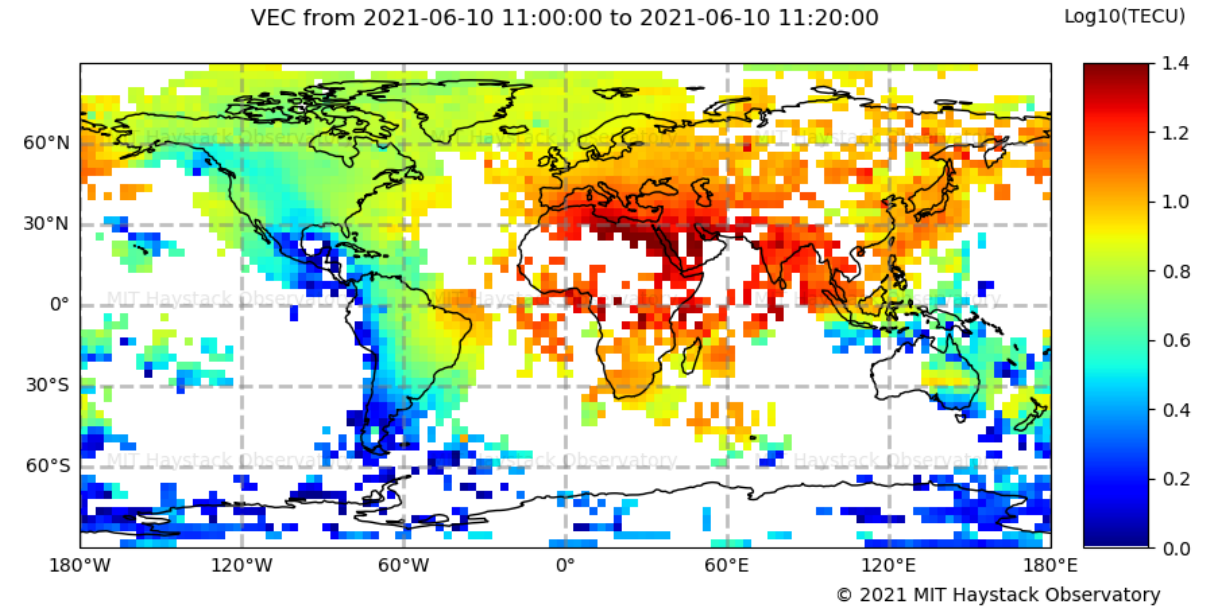
The Ionosphere

- It is a layer of the atmosphere that is made up of free electrons or ions due to the interactions of solar radiation and gases in the atmosphere
- The layer stretches from 75 km above the surface to 1000 km (remember that GPS satellites are about 20,000 km above the surface)
- The density of electrons is measured in TEC (total electron content). It is measured as the total amount in the ionospheric column above a square meter. 1 TEC is 10^{16} electrons



The Ionosphere

- The ionospheric component changes throughout the day (less at night) and throughout the year (greatest in Winter/Spring).
- Solar cycles affect the size of the ionospheric component and the geographic location (least in mid-latitudes).
- The ionosphere is dispersive, that is, the time delay depends on the frequency



The Ionosphere

- Because of the dispersive nature of the ionosphere, it is convenient to use the phase and group velocities to describe the impacts on the GPS observables.
- The GPS codes (C/A, P-code) travel at the group velocity, which slows the signal down (group delay)
- The carrier signals travel at the phase velocity, so the signal appears to speed up (phase advancement)
- Because of this, the ionospheric component has a positive sign on pseudorange code observations (because the signal is delayed) and a negative sign on phase observations (because the signal is advanced)

Ionospheric Delay

The ionospheric component (in meters) is given as the integrals of the electron content along the path:

$$\Delta I_{iono, ph} = -\frac{40.3}{f^2} \int N_e ds_0$$

$$\Delta I_{iono, gr} = \frac{40.3}{f^2} \int N_e ds_0$$

We usually express these equations in units of TEC along the slant (given as STEC):

$$I_f = \Delta I_{iono, gr}, f = \alpha_f STEC$$

$$\alpha_f = \frac{40.3 \cdot 10^{16}}{f^2} \frac{m}{TECU}$$

Note that this equation is only used to convert to TEC units. We are usually computing the ionospheric delay in meters since our goal is positioning.

Ionospheric Delay

$$I_f = \Delta I_{iono, gr, f} = \alpha_f STEC$$

$$\alpha_f = \frac{40.3 \cdot 10^{16}}{f^2} \frac{m}{TECU}$$

Our observation equations look like the following with the addition of the ionospheric terms:

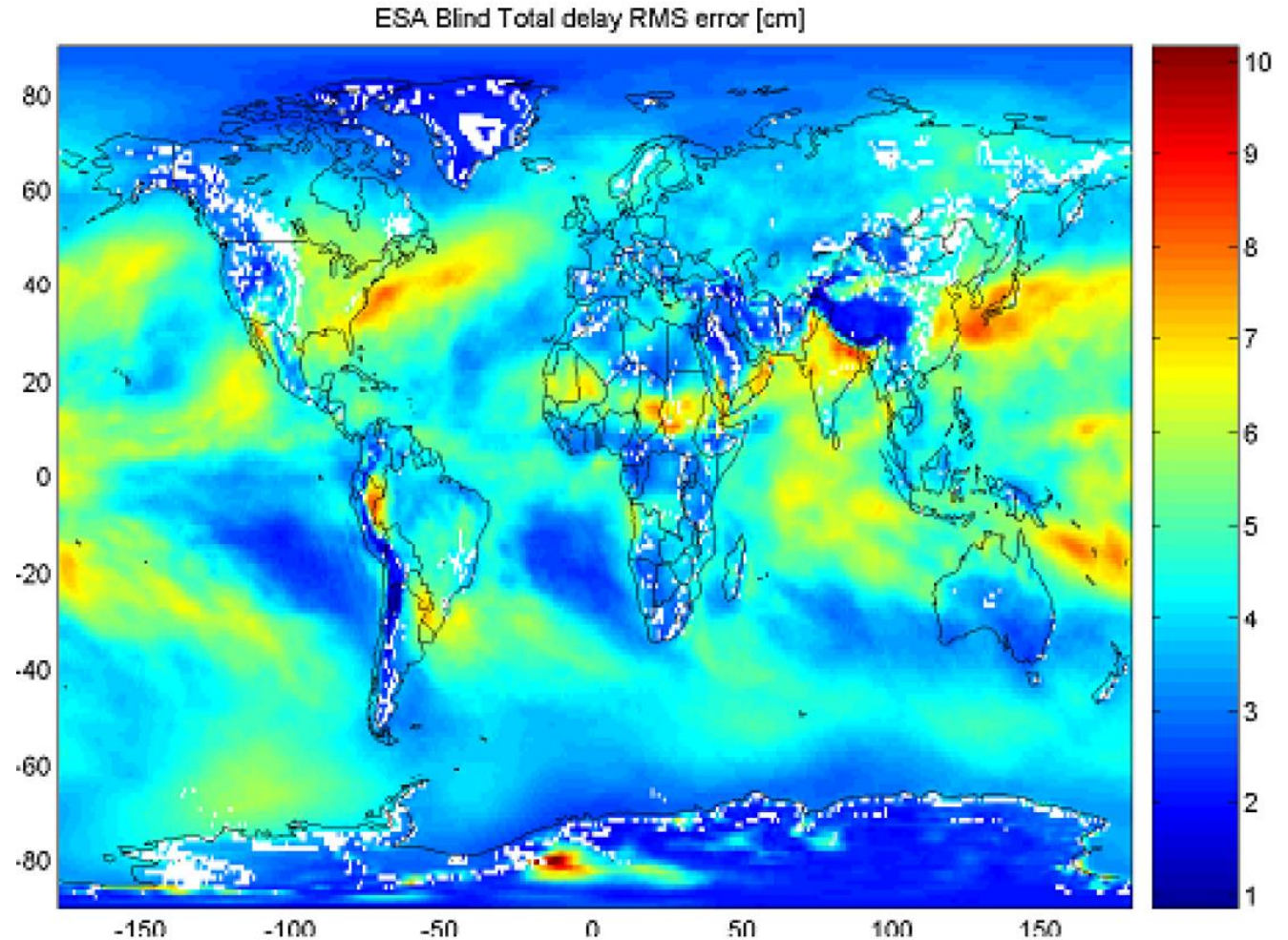
$$P_r^s(t) = \rho(t, t_s) + (\tau_r - \tau_s)c + I_r^s$$

$$L_r^s(t) = \rho(t, t_s) + (\tau_r - \tau_s)c - I_r^s + B_r^s$$

Remember that the ionospheric term is frequency dependent here, which we take advantage of later.

Troposphere

- The troposphere includes all the atmospheric material from the surface of the Earth to about 60 km.
- Unlike the ionosphere, the troposphere is non-dispersive for EM waves below 15 GHz, so both the carrier and phase delays will be the same and not dependent on frequency of the signal.



Troposphere

- We separate the tropospheric delay into two terms: the zenith dry/hydrostatic delay (**ZHD**) and the zenith wet delay (**ZWD**). The summation of these two terms is the zenith total delay (**ZTD**)
- The ZHD term is due to the total weight of the atmosphere above a point. It is predictable with knowledge of the temperature and pressure (about 90% of total variability).
- The ZWD term is due to variations in water vapor (i.e. clouds). This term can vary rapidly and is non-repeatable; it is much smaller in value than the ZHD term however not predictable.

Troposphere

We often will correct directly for the ZHD term. The Saastamoinen model is:

$$ZHD = \frac{0.00227768P_0}{1 - 0.00266 \cos(2\phi) - 0.00028h_{ref}}$$

Where ϕ is geodetic latitude, h_{ref} is the height of the receiver above the geoid, and P_0 is the surface pressure.

We then refer to the slant total delay term, which is given by mapping functions. A continued fraction is used here, the coefficients defined by the Vienna Mapping Function 1:

$$STD(\epsilon) = mf_h(\epsilon)ZHD + mf_w(\epsilon)ZWD \qquad mf(el) = \frac{1 + \frac{a}{1 + \frac{b}{c}}}{\sin el + \frac{a}{\sin el + \frac{b}{\sin el + c}}}.$$

Troposphere: Observation Equations

Adding onto the observation equations, we get:

$$P_r^s(t) = \rho(t, t_s) + (\tau_r - \tau_s)c + I_r^s + M_r^s ZTD_r^s$$

$$L_r^s(t) = \rho(t, t_s) + (\tau_r - \tau_s)c - I_r^s + B_r^s + M_r^s ZTD_r^s$$

In the above equations, you can use the hydrostatic delay or the total delay. You would generally subtract the SHD from the left-hand side and then solve for the SWD in the inversion

In GNSS meteorology, segregating the two tropospheric components is key.

Additional Observation Equation Terms

The GNSS observation equation consists of the true range from the satellite-to-receiver and a series of path delays and errors:

$$L_i = \underbrace{\rho}_{\text{True Range}} + \underbrace{c(\tau_r - \tau_s)}_{\text{Receiver \& Satellite Clock Errors}} + \underbrace{T}_{\text{Tropospheric Delay}} - \underbrace{\tilde{\alpha}_i(I + K_{21})}_{\text{Ionospheric Delay}} + \underbrace{B_i}_{\text{Differential Code Bias}} + \underbrace{\lambda_i w}_{\text{Phase Windup}} + \underbrace{m_i}_{\text{Phase Ambiguity}} + \underbrace{\epsilon_i}_{\text{Multipath}}$$

Phase Observation

Other terms are corrected outside of observation equation such as phase center offsets and relativistic terms

Precise Point Positioning and Relative Positioning

Relative Positioning

Relative positioning is helpful since it allows for the removal of common terms, simplifying the observation equation.

We start with the observation equation for two stations, A and B, and one satellite, j:

$$L_A^j = \rho_A^j + c\tau_A - c\tau^j + T_A^j - I_A^j + B_A^j$$

$$L_B^j = \rho_B^j + c\tau_B - c\tau^j + T_B^j - I_B^j + B_B^j$$

The difference is:

$$\Delta L_{AB}^j = L_A^j - L_B^j = (\rho_A^j - \rho_B^j) + c(\tau_A - \tau_B) + (T_A^j - T_B^j) - (I_A^j - I_B^j) + (B_A^j - B_B^j)$$

$$\Delta L_{AB}^j = \Delta \rho_{AB}^j + c\Delta \tau_{AB} + \Delta T_{AB}^j - \Delta I_{AB}^j + \Delta B_{AB}^j$$

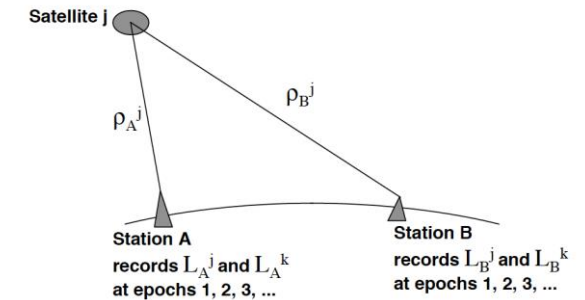


Figure 4: Single differencing geometry

Single difference equation
removes satellite clock biases!

Relative Positioning: Double-Difference

Next, we make another difference between two satellites, **j** and **k**, giving us the following equations:

$$\Delta L_{AB}^j = \Delta \rho_{AB}^j + c\Delta\tau_{AB} + \Delta T_{AB}^j - \Delta I_{AB}^j + \Delta B_{AB}^j$$

$$\Delta L_{AB}^k = \Delta \rho_{AB}^k + c\Delta\tau_{AB} + \Delta T_{AB}^k - \Delta I_{AB}^k + \Delta B_{AB}^k$$

$$\Delta L_{AB}^j - \Delta L_{AB}^k = \nabla \Delta \rho_{AB}^{jk} + \nabla \Delta T_{AB}^{jk} - \nabla \Delta I_{AB}^{jk} + \nabla \Delta B_{AB}^{jk}$$

Double difference equation removes receiver clock biases!

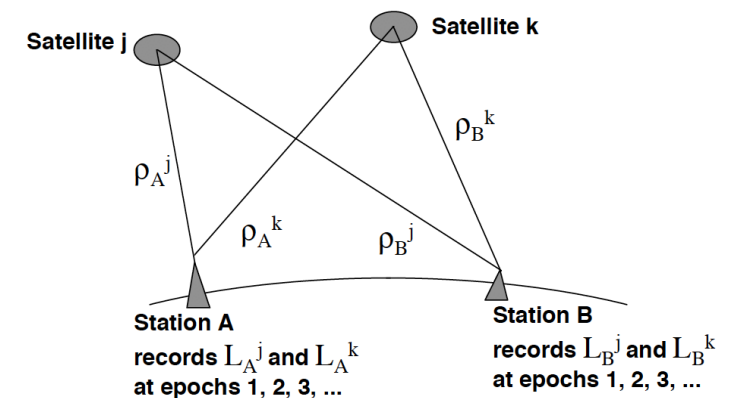


Figure 5: Double differencing geometry.

Relative Positioning: Double-Difference

The Ambiguity of the double difference solution is also an integer

$$\begin{aligned}\nabla\Delta B_{AB}^{jk} &= \Delta B_{AB}^j - \Delta B_{AB}^k \\ &= (B_A^j - B_B^j) - (B_A^k - B_B^k) \\ &= \lambda_0(\varphi_{0A} - \varphi_0^j - N_A^j) - \lambda_0(\varphi_{0B} - \varphi_0^j - N_B^j) - \lambda_0(\varphi_{0A} - \varphi_0^k - N_A^k) + \lambda_0(\varphi_{0B} - \varphi_0^k - N_B^k) \\ &= -\lambda_0(N_A^j - N_B^j - N_A^k + N_B^k) \\ &= -\lambda_0\nabla\Delta N_{AB}^{jk}\end{aligned}$$

Relative Positioning: Concepts

- In relative positioning, we use the double difference equations to compute baselines between a known “reference” station and a “rover” or many “rovers”. We also need to consider a reference satellite to ensure a complete solution
- If the baselines between the two stations are small, we can assume the tropospheric and ionospheric terms are identical for observations to the same satellite. This assumption is good to about 40 km, max of 100 km in arid regions.
- Relative positioning forms the basis for most GPS surveying techniques where a permanent GPS station or reference monument is used to compute baselines to a rover.

Relative Positioning: Concepts

If we have two receivers and three satellites, we have three possible double differences $(s-1)*(r-1)$

$$L_{AB}^{jk} = (L_A^j - L_B^j) - (L_A^k - L_B^k)$$

$$L_{AB}^{jl} = (L_A^j - L_B^j) - (L_A^l - L_B^l)$$

$$L_{AB}^{lk} = (L_A^l - L_B^l) - (L_A^k - L_B^k)$$

The generalized observation equation for the double difference assuming the reference and rover are close is:

$$L_{AB}^{jk} = \rho_{AB}^{jk} - \lambda_0 N_{AB}^{jk}$$

Given this equation, we have an independent parameter set of 6 to solve:

$$\{x_B, y_B, z_B, N_{AB}^{21}, N_{AB}^{23}, N_{AB}^{13}\}$$

Precise Point Positioning (PPP)

Relies heavily on products for clock biases, orbits, and earth parameters to solve for the position without the double difference or a minimum of reference stations.

The errors are generally greater in PPP, however, the improvements in products has made PPP comparable to RP.

It has the added advantage of computing positions in a global reference frame and therefore impacts to the reference station don't impact positioning accuracy (i.e. during earthquakes)

Equations are also less computationally intensive since fewer observation equations are employed

Linear Combinations of Phase and Pseudorange

The Ionosphere-Free Phase Combination

The effect of the first-order ionosphere term can be rather large and impact positions up to 50 meters. To remove this term, we use a linear combination of phase that will cancel out the first-order ionosphere. Let us start with the phase terms, starting with L_1 :

$$L_1 = \rho + \lambda_1 N_1 + I$$

In this case, we refer to the ionospheric correction for L_1 as I . For L_2 , we need to multiply by the frequencies since we have:

$$I = \frac{a}{f^2}$$

Therefore, the L_2 ionosphere term will be the L_2 term multiplied by the ratio of frequencies, q :

$$I_2 = \frac{f_1^2}{f_2^2} I = q^2 I$$

The Ionosphere-Free Phase Combination

We have the two general observation equations:

$$L_1 = \rho + \lambda_1 N_1 - I$$
$$L_2 = \rho + \lambda_2 N_2 - q^2 I$$

The linear combination of these two yields the following:

$$LC = \alpha L_1 + \beta L_2$$

The units on LC is meters here. We now need to find the values of the coefficients:

$$LC = \rho(\alpha + \beta) + \alpha \lambda_1 N_1 + \beta \lambda_2 N_2 - (\alpha + \beta q^2) I$$

We have the following two equations to get rid of the ionosphere:

$$\alpha + \beta = 1$$
$$\alpha + \beta q^2 = 0$$

The Ionosphere-Free Phase Combination

We initially solve for the beta term. We have the following equations and obtain beta:

$$\begin{aligned}\alpha + \beta &= 1 \\ \alpha + \beta q^2 &= 0 \\ 1 - \beta + \beta q^2 &= 0 \\ \beta(1 - q^2) &= 1\end{aligned}$$
$$\beta = \frac{1}{(1 - q^2)} = \frac{1}{\left(1 - \frac{f_1^2}{f_2^2}\right)} = \frac{f_2^2}{f_2^2 - f_1^2}$$

Alpha then becomes:

$$\alpha + \frac{f_2^2}{f_2^2 - f_1^2} = 1$$
$$\alpha = 1 - \frac{f_2^2}{f_2^2 - f_1^2} = \frac{f_2^2 - f_1^2}{f_2^2 - f_1^2} - \frac{f_2^2}{f_2^2 - f_1^2} = -\frac{f_1^2}{f_2^2 - f_1^2}$$

The Ionosphere-Free Phase Combination

Since f_1 is greater than f_2 , we will switch the signs around.

$$\alpha = \frac{f_1^2}{f_2^2 - f_1^2}, \quad \beta = -\frac{f_2^2}{f_2^2 - f_1^2}$$

So, the ionosphere-free combination becomes:

$$LC = \frac{f_1^2}{f_2^2 - f_1^2} L_1 - \frac{f_2^2}{f_2^2 - f_1^2} L_2$$

$$f_1 = 1575.42 \text{ MHz (154)}$$

$$f_2 = 1226.60 \text{ MHz (120)}$$

$$LC = 2.546L_1 - 1.546L_2$$

Note that while the combination was done in range, it holds the same constants in cycles.

The Geometry-Free Phase Combination

The ionosphere-free combination removed the dispersive terms. If we want to keep the dispersive terms and remove the other terms, we form the **geometry-free** combination.

If we converted phase to meters by multiplying by the wavelength, the geometry-free combination is simply:

$$LG = L_1 - L_2$$

If we are still in cycles, we use the gear ratio between L1 and L2, which is 60/77.

This combination is used for extracting the ionospheric component and for ambiguity resolution.

Melbourne-Wubbena WL

A linear combination used to obtain integer ambiguities is the Melbourne-Wubbena Wide-Lane. It is the wide-lane combination of phase and narrow-lane combination of code:

$$MWWL = N_2 - N_1 = (\phi_2 - \phi_1) - \frac{f_1 - f_2}{f_1 + f_2}(P_1 + P_2)$$

$$MWWL = N_2 - N_1 = (\phi_2 - \phi_1) - \frac{f_1 - f_2}{f_1 + f_2} \left(\frac{f_1}{c} P_1 + \frac{f_2}{c} P_2 \right)$$

The MW-WL has an effective wavelength of 86.2 cm, much larger than L_1 or L_2 alone. This spread makes the resolution of integer ambiguities easier.

General Precise Positioning Procedure

Precise Positioning

We start by removing terms that are known or can be estimated. For example, the satellite clock and hydrostatic delay. We subtract these terms from the phase observations to form the **pre-fit residuals**.

Ambiguities can be solved next or in the linear observation model. For post processing, the ambiguity state can be determined across the whole day through iteration. FCBs need to also be solved for PPP.

Ambiguities will also get subtracted from the pre-fit residuals. From here, the station position, the receiver clock bias, and the wet delay are jointly estimated

(https://gssc.esa.int/navipedia/index.php/Linear_observation_model_for_PPP)

Precise Positioning in GipsyX

Start with Raw 30-Hour GNSS Data

1. Ionosphere-Free Linear Combination (Filters out ionospheric delays)
2. Melbourne-Wübbena Combo + JPL WLPB Table (Resolves wide-lane integer)
3. Square-Root Information Filter (SRIF) (Saves wide-lane integers as tight constraints)
4. Bi-Directional Smoothing (Smooth narrow-lane float values globally over the 30-hr arc)

Final Daily 24-Hour Integer-Fixed Solution

This process is the start of obtaining a single time series point.

First Jupyter Notebook

First Jupyter Notebook

The focus of this book is on loading some raw GNSS data and interacting with it by performing some basic linear combinations, doing a simple point positioning, computing DOP, and determining the ionospheric/tropospheric components.

Additional tasks you can do are determining ambiguities by forming linear combinations of observables, mapping changes in ionospheric TEC, obtaining orbital information from the sdk tools.

Reference Frames

Reference Frames

- The geoid is the equal-potential surface of the Earth, that is along this surface the gravity field is the same. The geoid gives a rough approximation to the surface of the oceans under only the influence of Earth's gravity
- An ellipsoid is a mathematically defined surface that attempts to mimic the geoid.
- These ellipsoids are defined by their semi major and minor axes; the Earth is approximated as an oblate spheroid due to the rotation of the Earth. The radius at the poles is about 20 km less than at the equator.
- The WGS84 is one of the most common ellipsoids currently used, defined with an equatorial radius of 6378137 m and a flattening of roughly $1/298.257$

Reference Frames

- The International Terrestrial Reference Frame is an Earth centered, Earth fixed (ECEF) reference frame that appropriately describes deformations of the Earth as it pertains to locations on the surface of the Earth.
- A suite of techniques are employed to define the ITRF, including GNSS, VLBI, DORIS, and SLR
- The ITRF attempts to as accurately as possible define the geocenter of the Earth such that observations on or around it are not impacted by variations in the geocenter or in the axis of rotation.
- The ITRF attempts to mimic the WGS84 system closely, and at present is different by a few cms

Local Reference Frames

- Individual countries or states will often employ a local reference ellipsoid or datum to better describe position. In the US, the North American Datum 27 or 83 are such systems
- The Stable North America Reference Frame (SNARF) is another example of a local reference frame used in GPS
- There are also state plane systems used as well.

GNSS Time Series

GNSS Time Series Providers and Codes

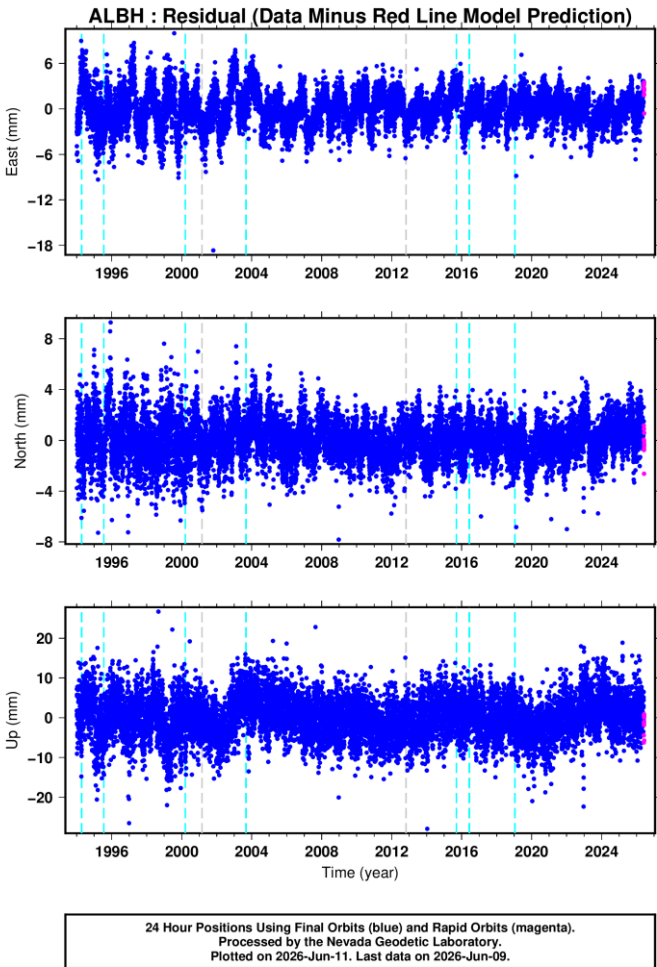
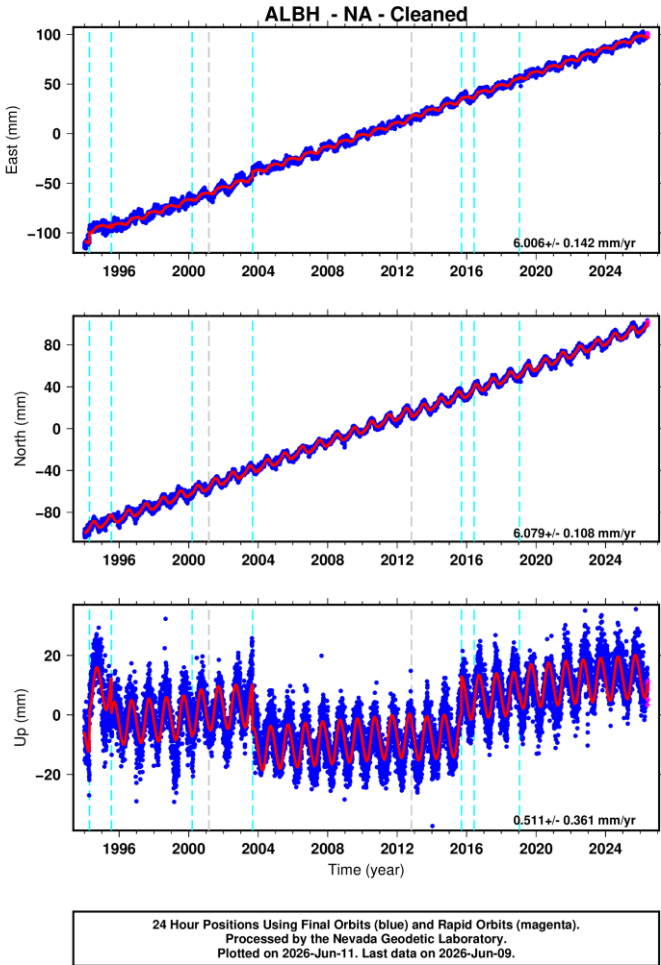
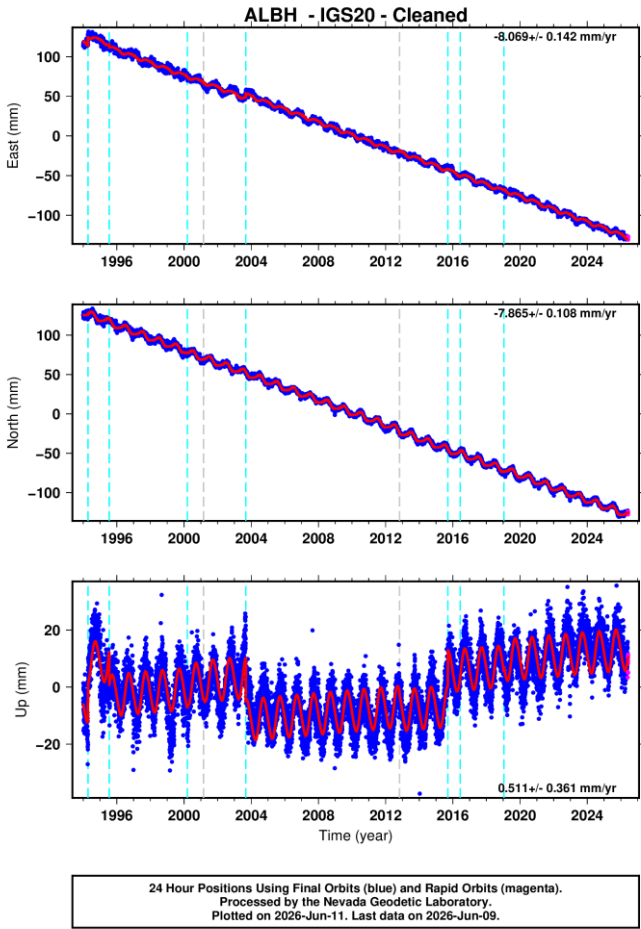
- The first half we covered how a GNSS position at a single epoch would be determined. While one could write their own code, there are many that are easily available (or the solutions are published).
- **Codes:** GAMIT/GlobK, GipsyX, Bernese, Pride PPP-AR
- **Public Time Series Providers:** UNR, SOPAC, JPL, PANGA, EarthScope
- While many of these providers use the same software, different processing choices lead to different solutions. It's important to understand the nuances.

GNSS Time Series Considerations

- Each time series provider will generate several different products.
- The highest level will generally be in ITRF (converted to local north, east, up) with minimal corrections outside of the following:
 - Solid Earth tides
 - Ocean tidal loading
 - Equipment change jumps
- After this, time series products will have higher-order corrections: detrending, fitting model terms, Euler pole correction, regional filtering, etc.

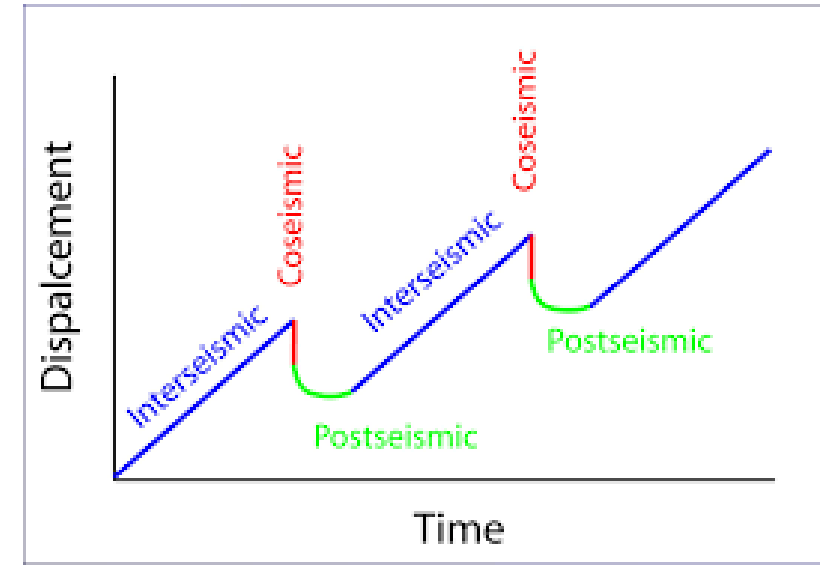
IGS20 Original
IGS20 Cleaned
NA Plate Fixed
NA Plate Fixed Cleaned
MIDAS Detrended
Detrended and Cleaned
Residual To Red Line Model
Rapid 5 Minute
Quality Assurance
Loading Predictions

UNR Example Time Series



Importance of Time Series Modeling

- By breaking down our GNSS time series into a series of components, we enable a slew of scientific and societal applications.
- The long-term trend gives us the interseismic velocity which describes plate tectonics at the surface.
- Coseismic offsets and postseismic relaxation tells us about moment release on active faults.



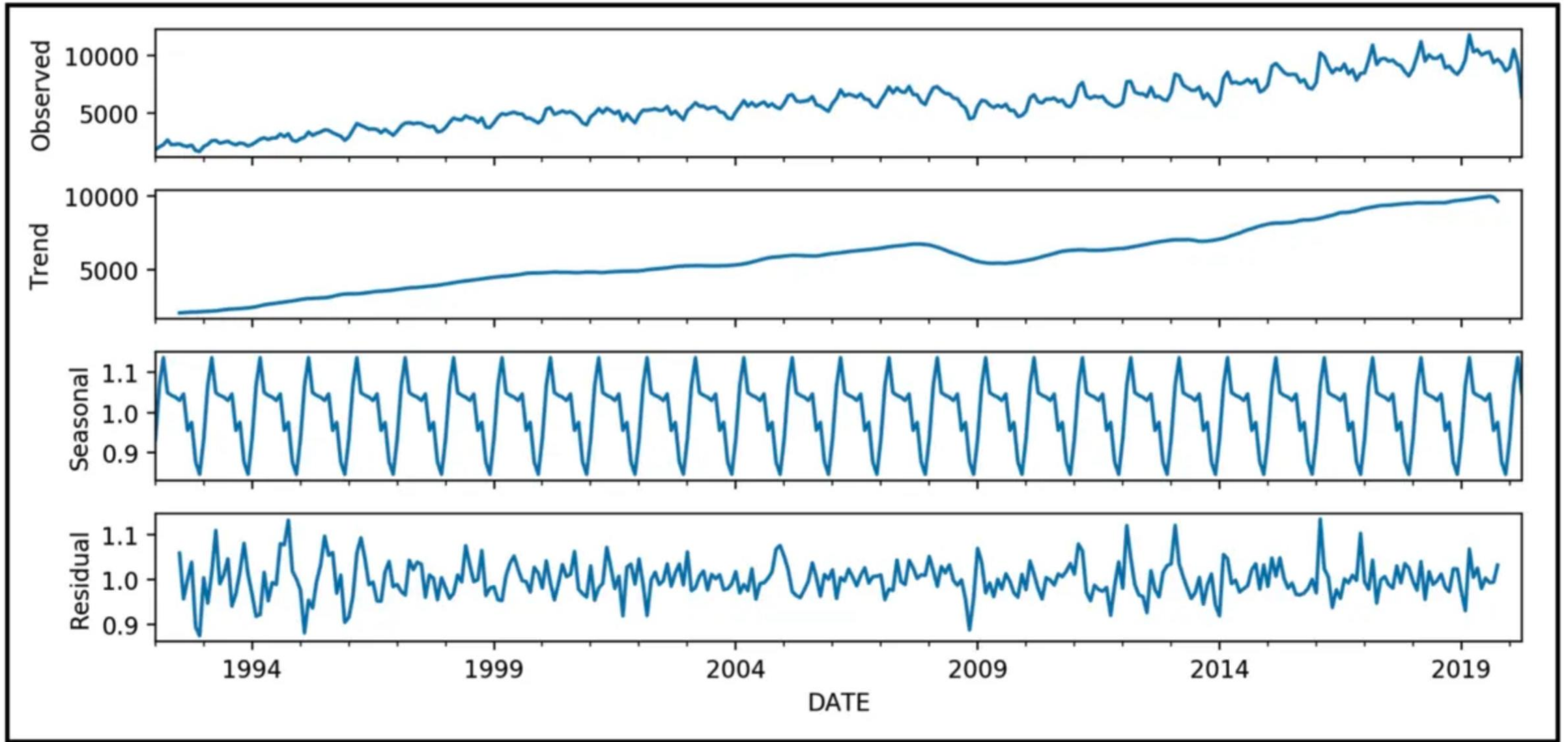
Importance of Time Series Modeling

- The seasonal terms tell us about the shallow structure, mostly driven by hydrologic effects.
- Transient terms tell us about slow-slip, volcanic deformation, or other types of deformation.
- All of these terms together help to create trajectory models that inform of new reference frame realizations.

Time Series Analysis

A major part of understanding a time series is breaking it down into its constituent parts, a process known as "decomposition." A time series is typically modeled as a combination of four elements:

- **Trend:** The long-term progression of the series (either upward, downward, or stable).
- **Seasonality:** Regular, predictable fluctuations that repeat over a fixed period (e.g., daily temperature drops at night, or retail sales spiking every December).
- **Cyclicity:** Rises and falls that are not of a fixed, predictable length, often driven by macroeconomic factors like recessions.
- **Irregularity (Noise):** The random, unpredictable variations left over after extracting the trend, seasonality, and cycles.



Time Series Analysis

The Principle of Superposition

- In time series analysis, the principle of superposition states that a complex observed signal is simply the algebraic sum of multiple, simpler underlying signals occurring simultaneously.
- When you look at a raw, messy time series plot, you are almost never looking at a single, isolated process. Instead, you are looking at the net result of many different forces acting together at the exact same time.

Time Series Analysis

The Mathematical Representation of Superposition

- Mathematically, if you have a set of independent signals, let's call them $x_1(t)$, $x_2(t)$, up to $x_n(t)$, the observed time series $y(t)$ is their sum. We can represent this using the following equation:

$$y(t) = \sum_{i=1}^n x_i(t)$$

- This additive property is the foundation of linear systems. If a system is linear, the output produced by a combination of inputs is exactly equal to the sum of the outputs that would be produced by each input individually.

Makeup of GPS Time Series

A general form for the GPS time series is given below. It contains a slope (b; velocity), seasonal terms (the sines and cosines), Heaviside step functions (for jumps, coseismic displacements), and decay terms (for post-seismic relaxation).

$$y(t_i) = a + bt_i + c\sin(2\pi t_i) + d\cos(2\pi t_i) + e\sin(4\pi t_i) + f\cos(4\pi t_i) \\ + \sum_{j=1}^{n_g} g_j H(t_i - T_{g_j}) + \sum_{j=1}^{n_h} h_j H(t_i - T_{h_j}) t_i \\ + \sum_{j=1}^{n_k} k_j e^{\left[1 - \left(\frac{t_i - T_{k_j}}{\tau_j}\right)\right]} H(t_i - T_{k_j}) + \varepsilon_i. \quad (40)$$

or

$$\sum_{j=1}^{n_k} k_j \log\left(1 + \frac{t_i - T_{k_j}}{\tau_j}\right) H(t_i - T_{k_j}).$$

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \varepsilon \quad (42)$$

where $\mathbf{A}$ is the design matrix and $\mathbf{x}$ is the parameter vector,

$$\mathbf{x} = (a \ b \ c \ d \ e \ f \ g \ h \ k)^T. \quad (43)$$

Inversion is implemented with a constrained weighted least squares algorithm (19)

$$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{l}; \quad \hat{\Sigma}_{\hat{\mathbf{x}}} = \hat{\sigma}_0^2 (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1}; \quad \hat{\sigma}_0^2 = \frac{\hat{\varepsilon}^T \mathbf{P} \hat{\varepsilon}}{n - u}. \quad (44)$$

Interseismic Velocity

The interseismic velocity is the long term trend of the time series and gives the motion of the tectonic plates.

If this velocity is in ITRF, we will need to rotate it into the local frame, which for us is the North American frame.

This rotation is performed with an Euler pole between NA and ITRF. The predicted motions from the Euler pole are evaluated at the location of the station, which is then subtracted from our ITRF velocities.

I won't go over the math here, but it is in the second Jupyter notebook.

Anatomy of Seasonal Variations

$$c \sin(2\pi t_i) + d \cos(2\pi t_i) + e \sin(4\pi t_i) + f \cos(4\pi t_i)$$

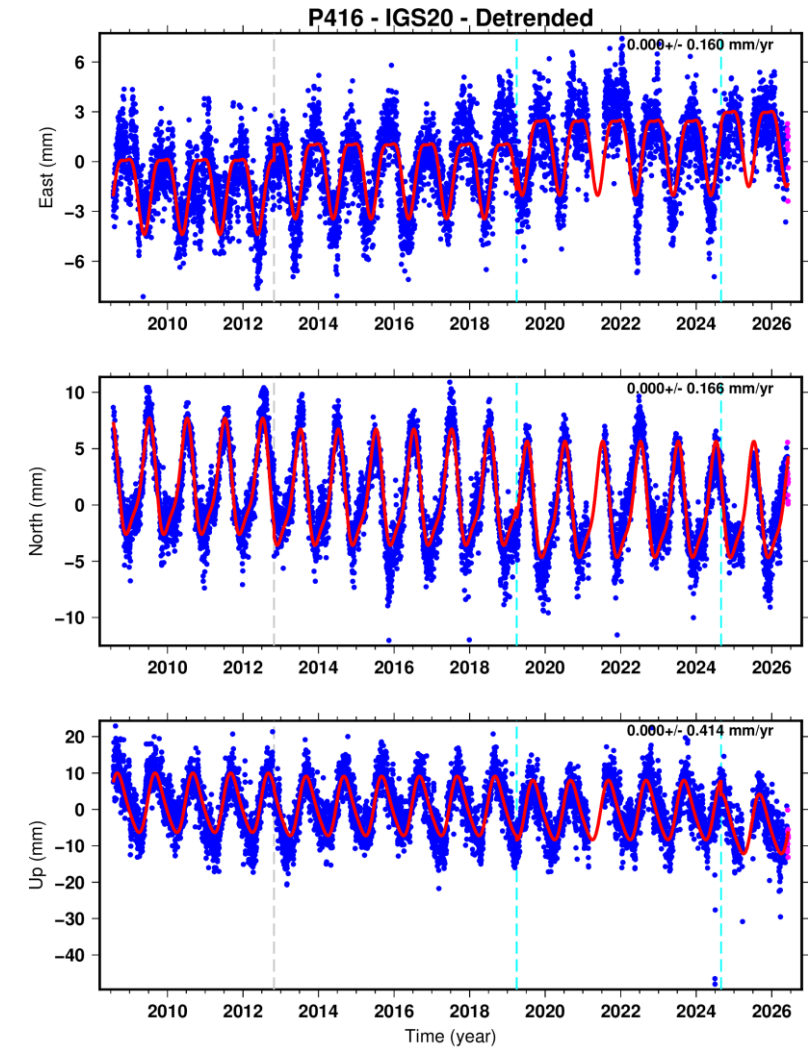
Table 3. Contributions of Geophysical Sources and Model Errors to the Observed Annual Vertical Variations in Site Positions

Sources	Range of Effects, mm
Pole tide	~4
Ocean tide	~0.1
Atmospheric mass	~4
Nontidal ocean mass	2–3
Snow mass	3–5
Soil moisture	2–7
Bedrock thermal expansion	~0.5
Errors in orbit, phase Center, and troposphere models	No quantitative results yet
Error in network adjustment ^a	~0.7
Differences from different software	~2–3; 5–7 at some sites

^aThe value is network-dependent.

Equations for pole tide variations – note x_p and y_p are EOPs defined by IERS (theta latitude, lambda is longitude):

$$\begin{aligned} d\lambda &= 9.0 \cos \theta (x_p \sin \lambda + y_p \cos \lambda) \\ d\theta &= -9.0 \cos 2\theta (x_p \cos \lambda - y_p \sin \lambda) \\ dr &= -32.0 \sin 2\theta (x_p \cos \lambda - y_p \sin \lambda), \end{aligned}$$



Jumps or Offsets

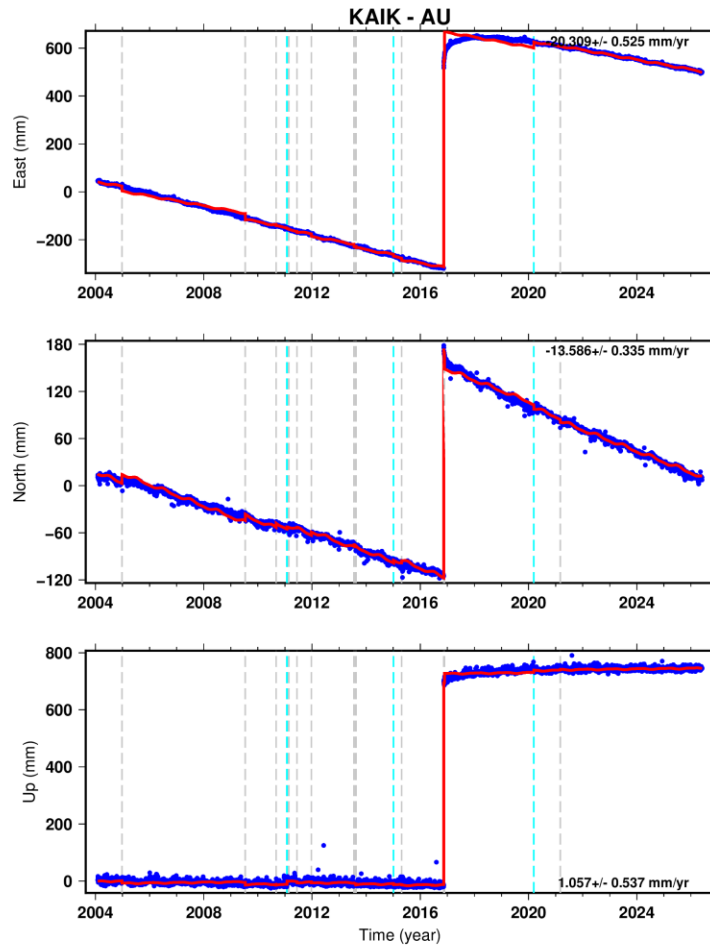
Each time series should be corrected for offsets so the interseismic velocity field is unbiased. There are two kinds of jumps: tectonic and non-tectonic.

Tectonic jumps are associated with earthquakes or slow slip events.

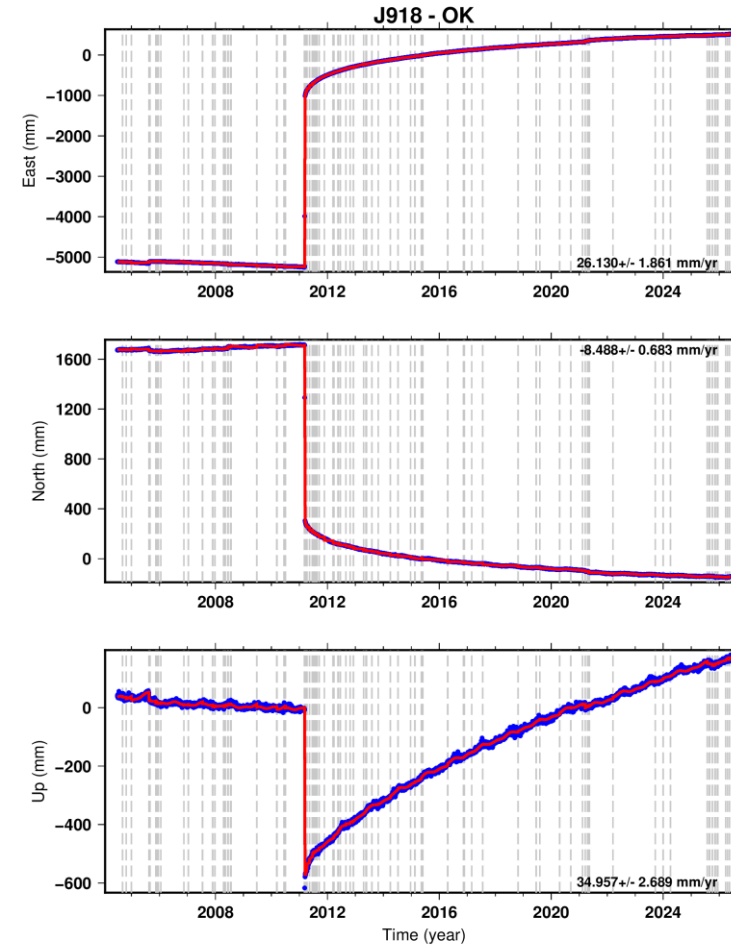
Non-tectonic jumps are generally associated with equipment changes or processing discontinuities (i.e., not converting to new ITRF). Instrument damage can also cause non-tectonic jumps.

$$+ \sum_{j=1}^{n_g} g_j H(t_i - T_{gj}) + \sum_{j=1}^{n_h} h_j H(t_i - T_{hj})$$

Jumps or Offsets



24 Hour Positions Using Final Orbits (blue) and Rapid Orbits (magenta).
Processed by the Nevada Geodetic Laboratory.
Plotted on 2026-Jun-11. Last data on 2026-May-30.



24 Hour Positions Using Final Orbits (blue) and Rapid Orbits (magenta).
Processed by the Nevada Geodetic Laboratory.
Plotted on 2026-Jun-11. Last data on 2026-May-30.

Postseismic Deformation

Postseismic deformation requires careful investigation. It is common to use either a log or exponential decay term with some associated decay constant.

$$+ \sum_{j=1}^{n_k} k_j e^{\left[1 - \left(\frac{t_i - T_{kj}}{\tau_j}\right)\right]} H(t_i - T_{kj})$$

This is a non-linear inverse problem. We generally will grid search over a range of decay terms while solving all of the other model terms. The lowest misfit is the chosen decay term.

$$\sum_{j=1}^{n_k} k_j \log\left(1 + \frac{t_i - T_{kj}}{\tau_j}\right) H(t_i - T_{kj}).$$

One could also do a PCA analysis across a network, find the decay PC, then fit a decay model to it.

Residuals

After we fit our model to the time series, we will theoretically want a set of residuals that show no biases and have an even variance about the mean.

Unfit model terms, usually in the form of unmodeled earthquakes or transient deformation events, will show up in the residual time series.

If you are working on slow slip events, you can work from this residual velocity field if any of the other model terms are not in conflict (seasonal terms at the same cadence of slow slip, earthquakes at the same time, etc.)

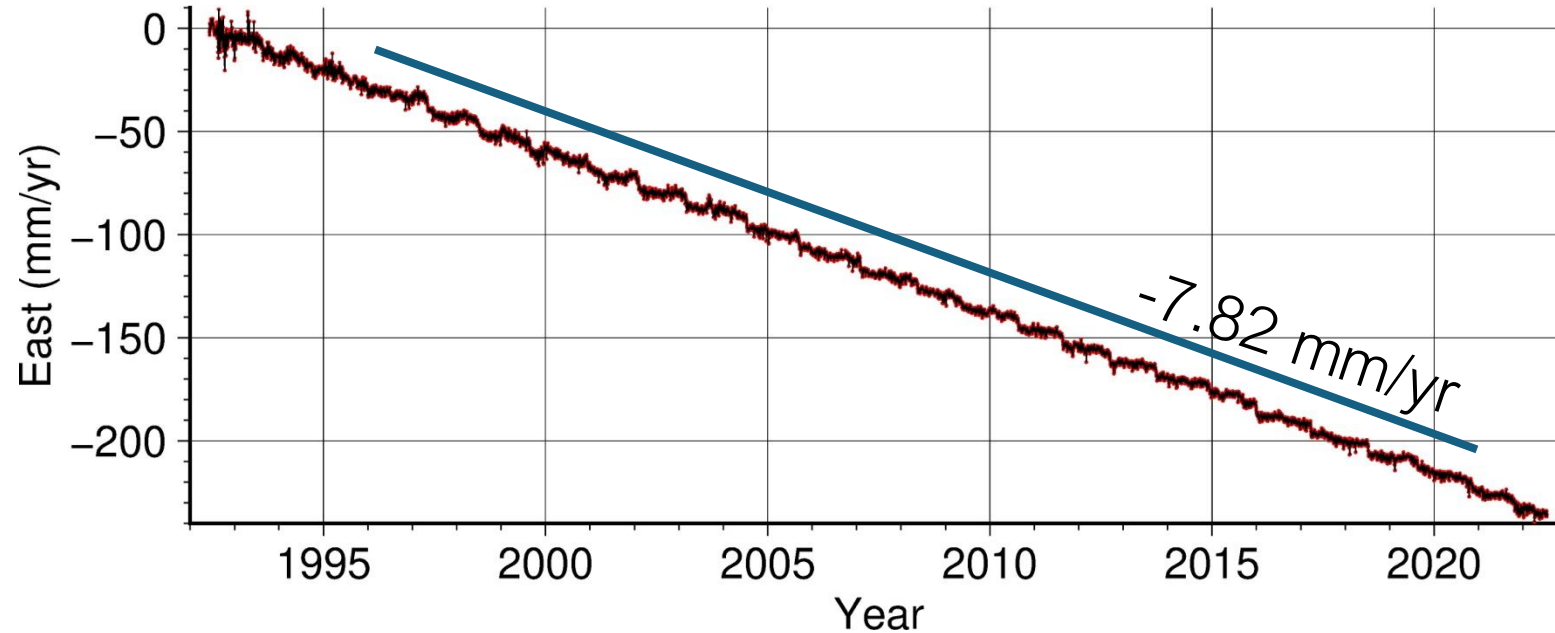
Slow-Slip Events

In Cascadia, slow-slip events were discovered in the early 2000s when it was noticed that GPS time series exhibited reversals at the same time as non-volcanic tremor. These events were termed 'episodic tremor and slip', or **ETS** for short. We also use the shorthand **SSE** for 'slow-slip event'. In this workshop, we aren't dealing with the tremor, so we will refer to SSEs.

In our residual time series, an SSE will show up as a reversal of the interseismic velocity over days to weeks (or longer in some areas).

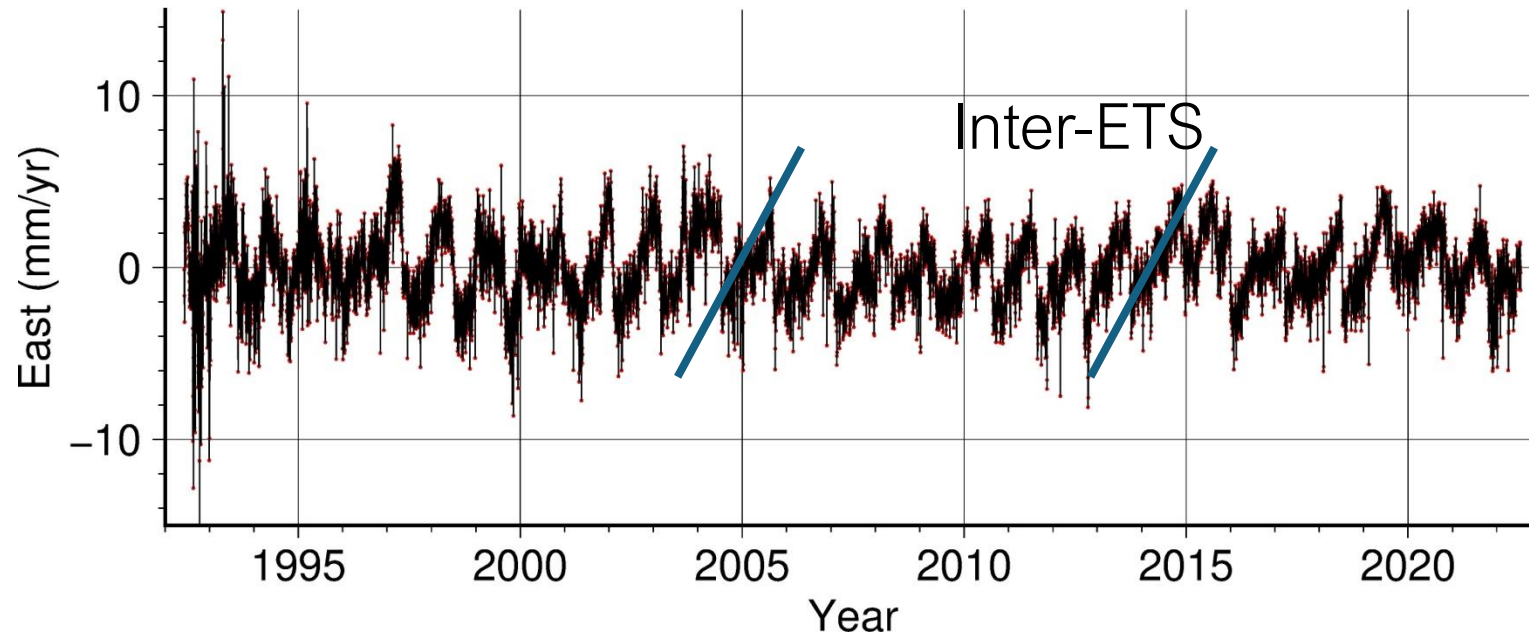
The period between SSEs will have a different slope from the total slope of the full time series. We term this rate as the 'inter-ETS' rate.

An Example Time Series - ALBH



On raw GNSS time series, the most dominant feature is the long-term trend, or interseismic motion.

Transients are barely visible at this scale.



After detrending, the transient behavior is revealed.

The velocity between slow-slip events, inter-ETS rate, is consistent at different times and the slow-slip events vary in size.

Second Jupyter Notebook

Second Jupyter Notebook

The focus of this book is on loading GNSS time series and modeling them. In this, we will decompose a single time series into its constituent parts, obtain offsets for slow slip events, estimate interseismic velocities across the network, perform Euler pole rotations, and generate the velocities necessary for Day 2.

Crescent Transient Deformation Workshop

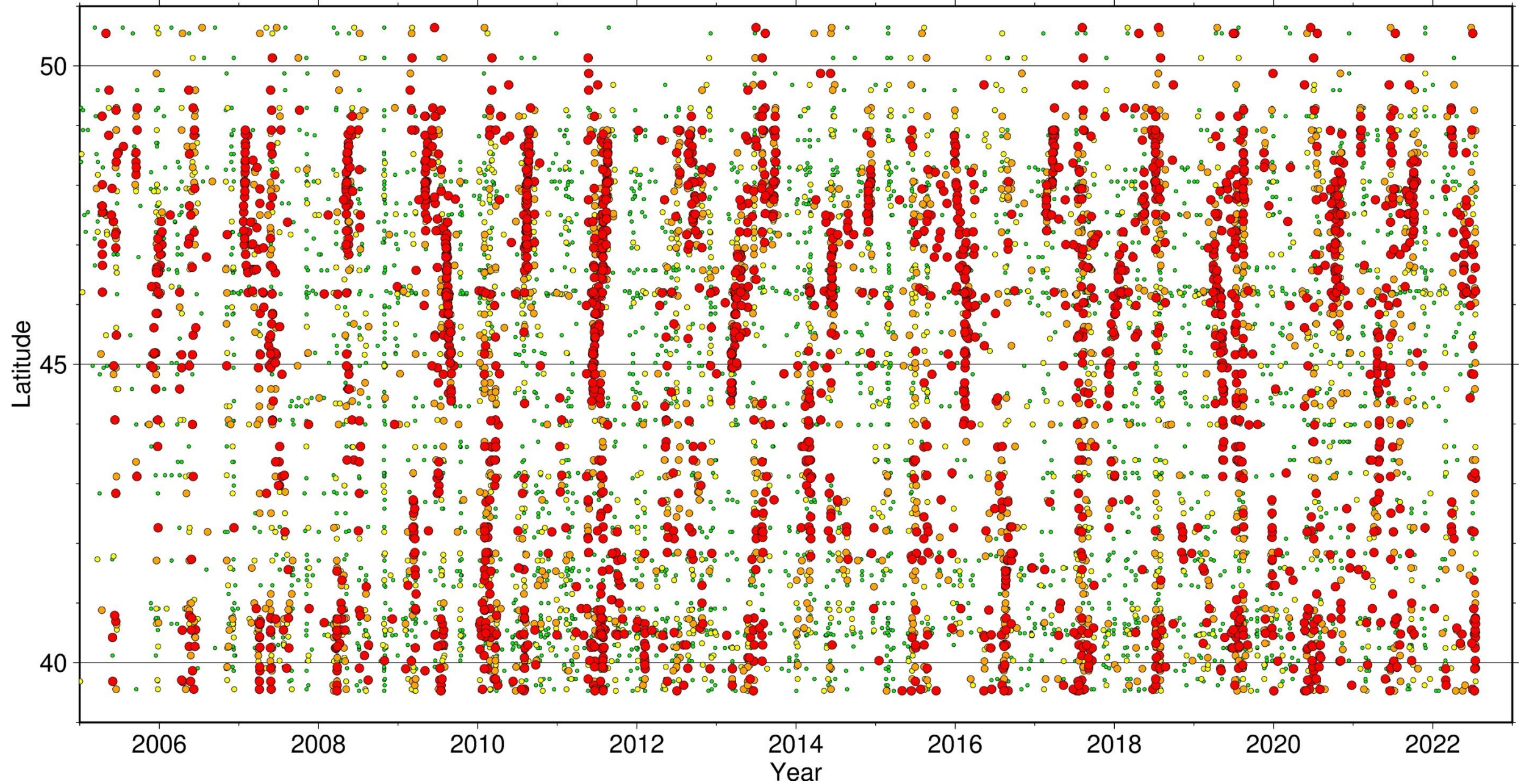
Transient Deformation Workshop

Many groups have developed methodologies for detecting SSEs and other transient deformation events, but each algorithm has caveats and no formal comparison between different algorithms and catalogs has been performed.

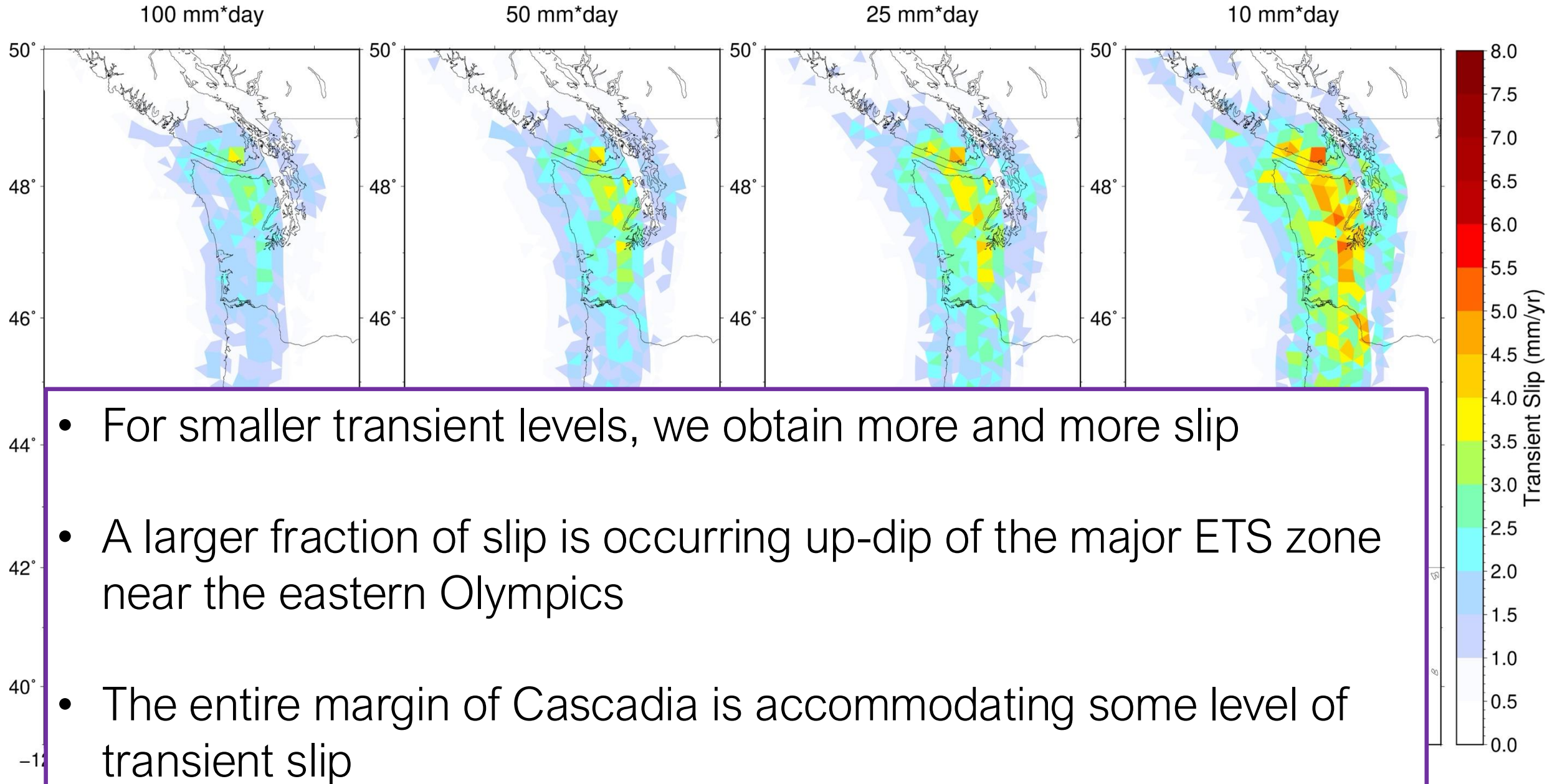
For C3S, we want to develop a robust and complete catalog of transient deformation throughout Cascadia using community developed algorithms.

This exercise will be going on asynchronously over the next 6 months to a year, and groups will submit their catalogs and codes for comparison.

Example Catalog from My Algorithm

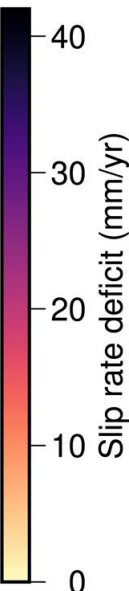
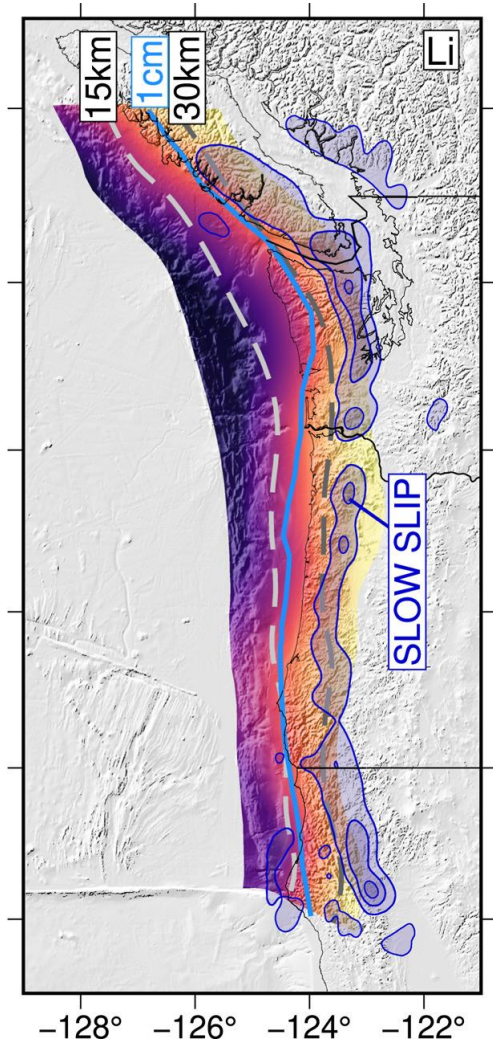
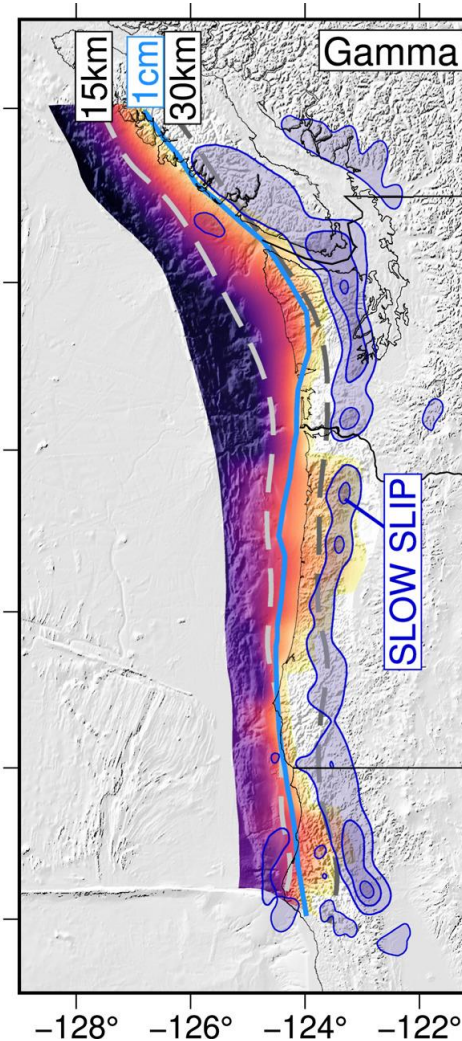
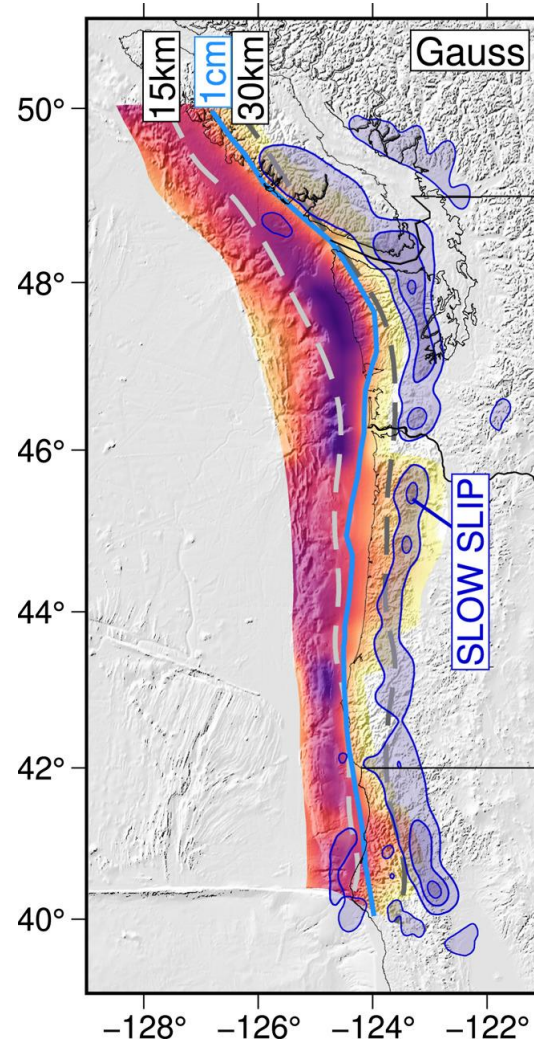
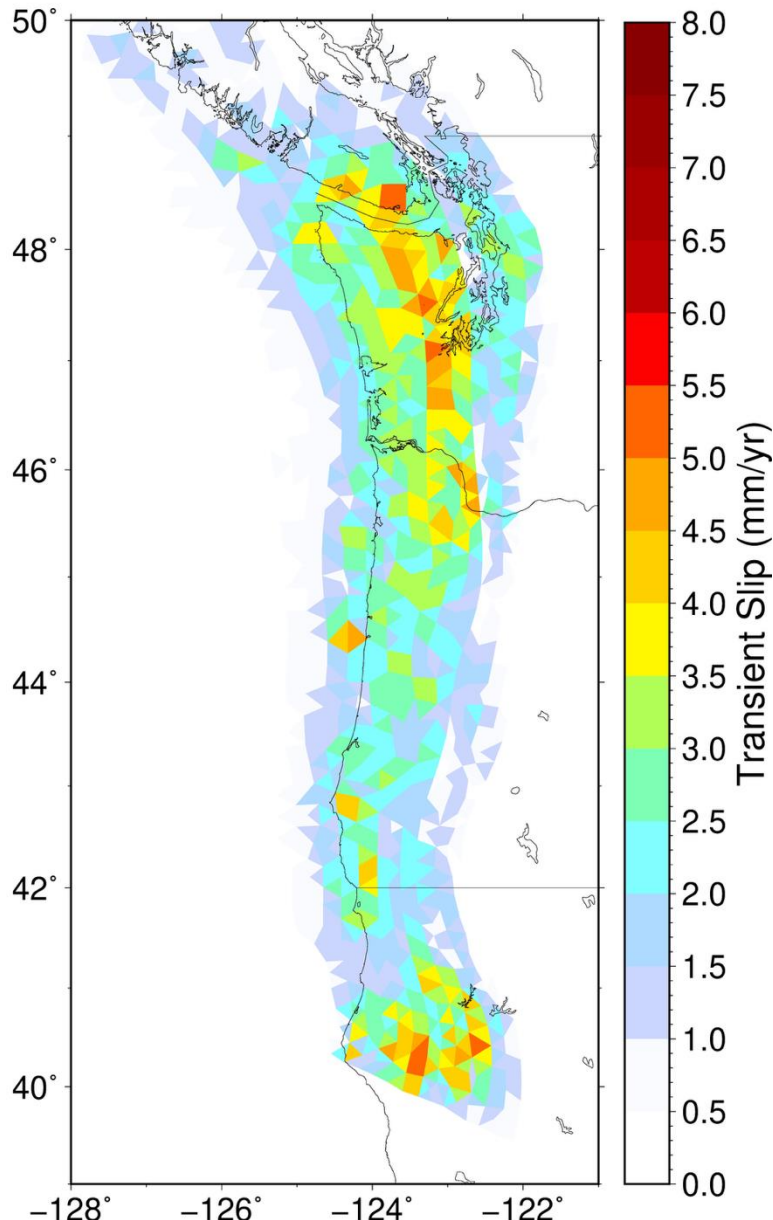


Implications of Better Resolved Transient Behavior



Implications of Better Resolved Transient Catalog

10 mm*day



Melgar et al. (2022); Schmalzle et al. (2014); Li et al. (2018)

How to Participate?

Send an email to me (crowell.97@osu.edu) or Margarita Solares-Colon (solares-colon.1@osu.edu) expressing your interest.

Utilize the Jupyter notebook and requirements document for understanding required parameters to submit. These are being finalized now, should be fully available in the next few weeks.

The second Jupyter notebook used today uses the same data format as the transient exercise, so you can use the initial parts as a kernel for algorithm development.